

Steve's Army -- Architecture Documentation

Mod ID: `steves_army`

Minecraft: 1.20.1 Forge (Parchment mappings)

Entrypoint: `StevesArmyMod.java`

Dependency: TaCZ (optional, via reflection)

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1. Module Overview & Goal Priority

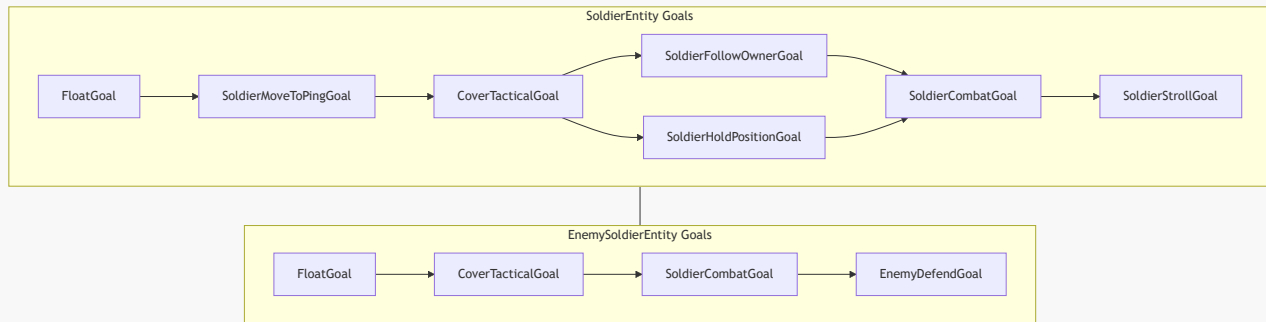
Goal Registration Order

Friendly `SoldierEntity`

Priority	Goal
0	FloatGoal
1	SoldierMoveToPingGoal (ping-driven movement)
2	CoverTacticalGoal (cover seeking/staying)
3	SoldierFollowOwnerGoal (FOLLOW mode)
3	SoldierHoldPositionGoal (HOLD mode)
4	SoldierCombatGoal (targeting, shooting, suppression)
5	SoldierStrollGoal (idle wander)
6	LookAtPlayerGoal
7	RandomLookAroundGoal

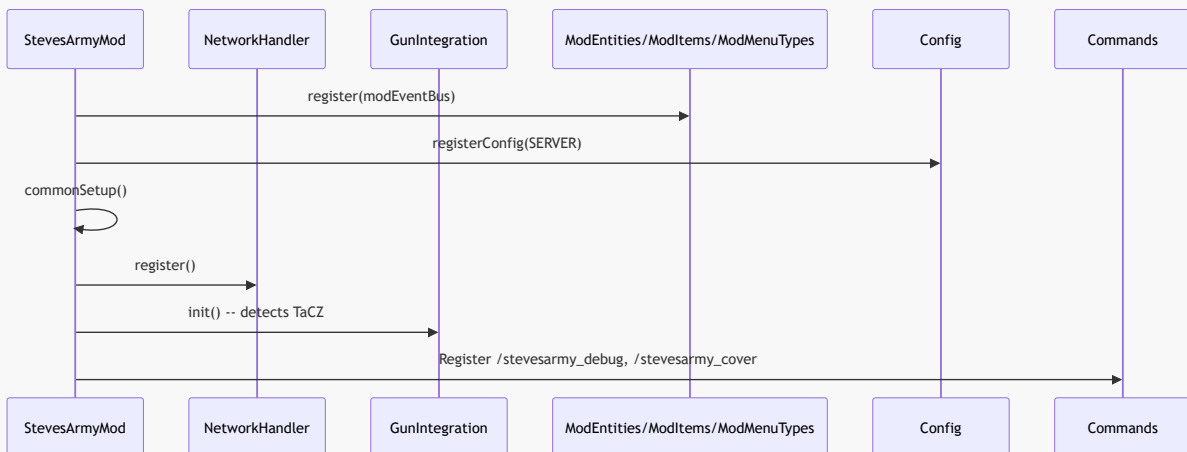
Enemy `EnemySoldierEntity`

Priority	Goal
0	FloatGoal
1	CoverTacticalGoal (cover seeking/staying)
2	SoldierCombatGoal (targeting, shooting, suppression)
3	EnemyDefendGoal (defend spawn position)
4	LookAtPlayerGoal
5	RandomLookAroundGoal



Key: **SoldierMoveToPingGoal** has higher priority than cover. A soldier with a valid ping move target will **not** seek cover until the ping expires.

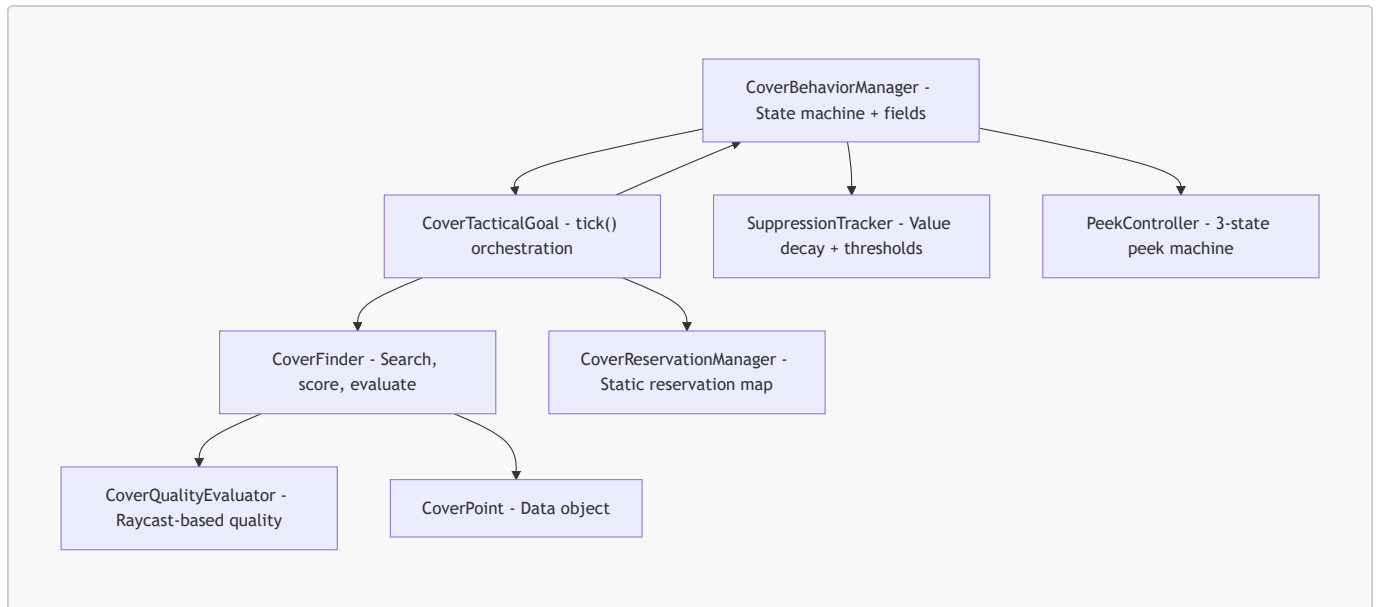
Entrypoint Flow



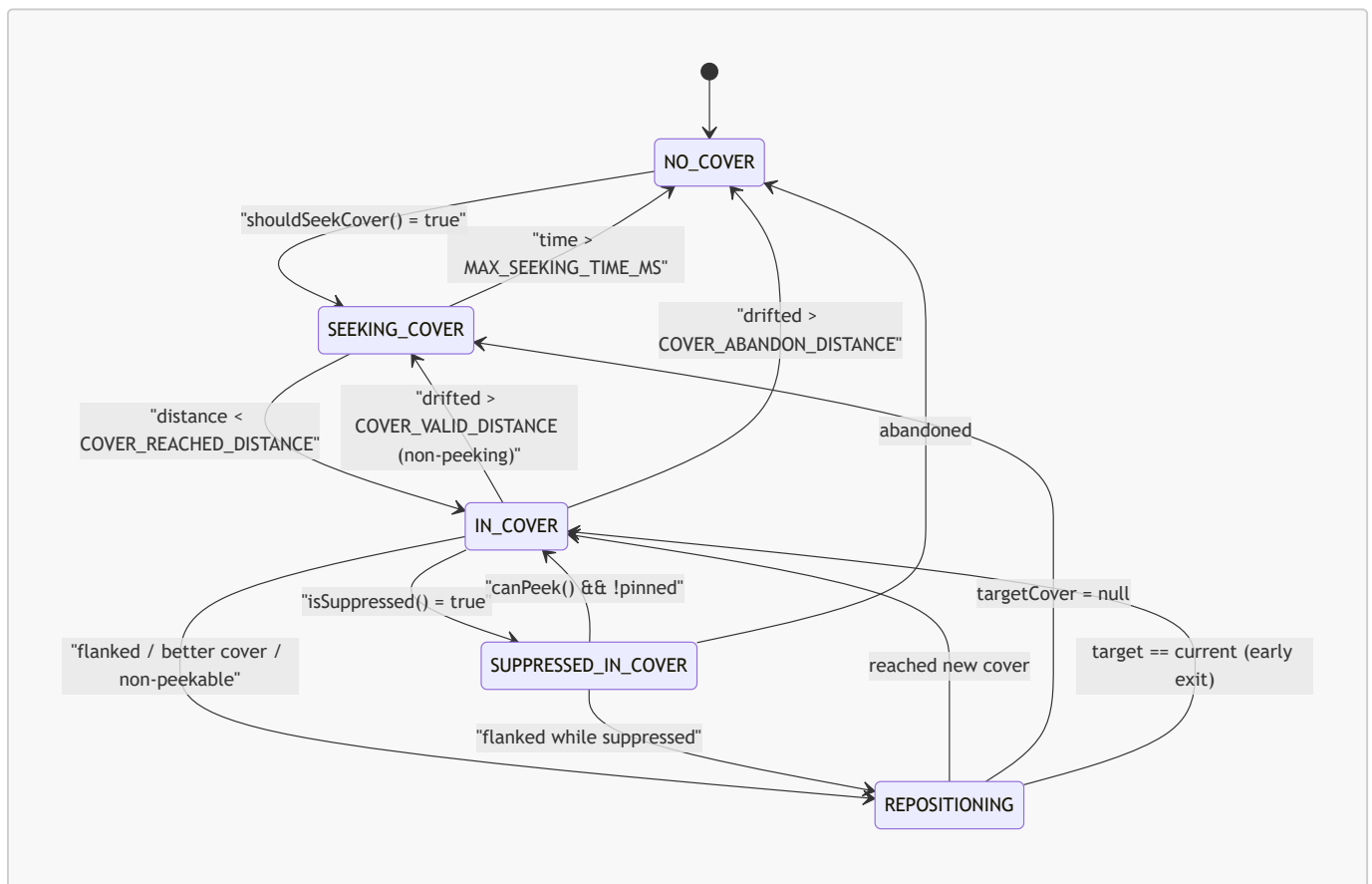
2. Cover System

Package: `com.stevesarmy.combat.cover`

The cover system is the core tactical AI module. It consists of several single-responsibility classes.



2.1 Cover State Machine



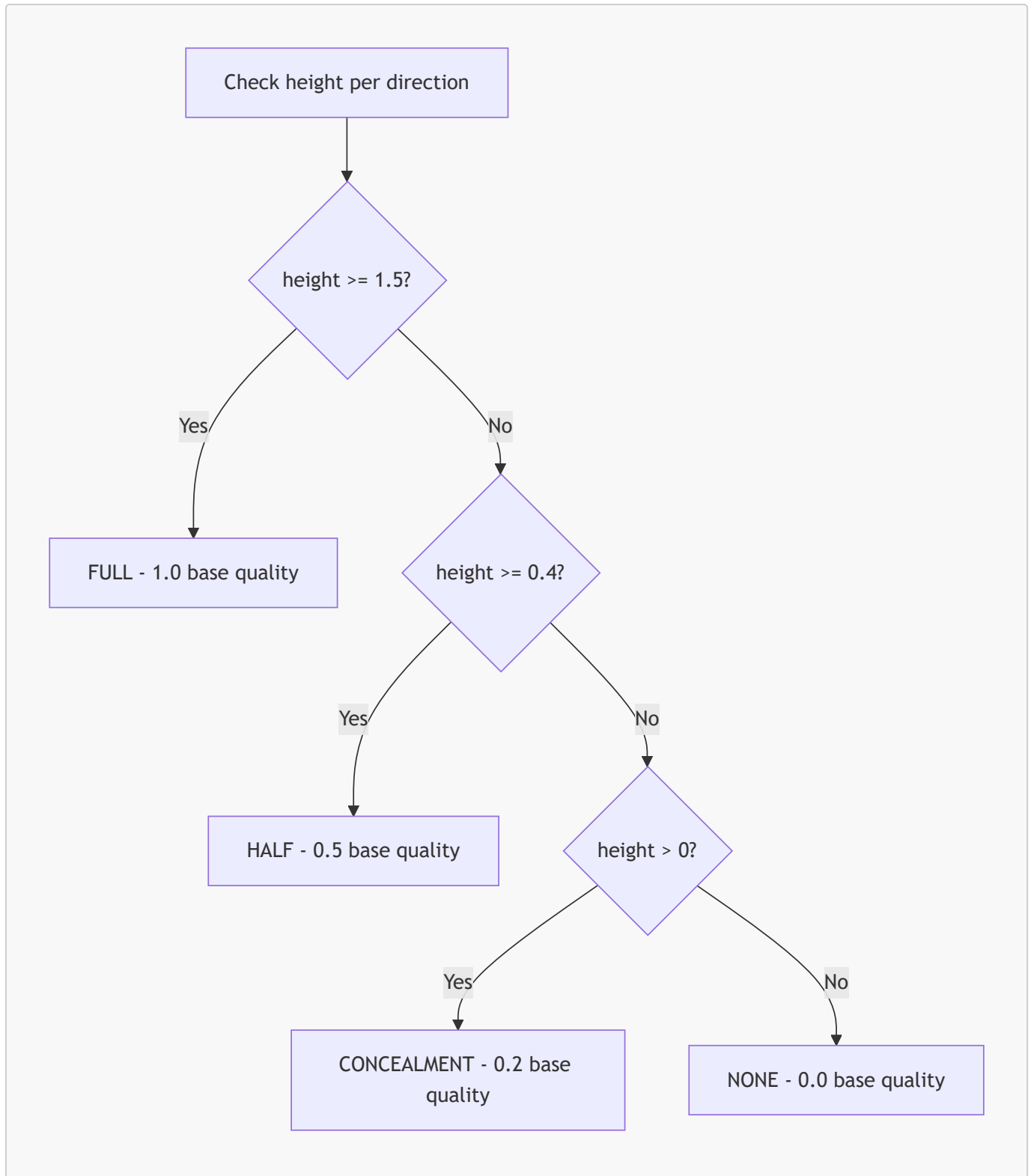
2.2 Cover Scoring Weights

Calculated in `CoverFinder.calculateThreatAwareScore()`

Component	Weight	Description
PRIMARY_PROTECTION_WEIGHT	0.25	Protection from primary threat direction
FIRING_QUALITY_WEIGHT	0.20	Ability to fire toward threat while protected
FLANKING_PROTECTION_WEIGHT	0.15	Protection from multiple threats

Component	Weight	Description
PEEK_ANGLE_WEIGHT	0.15	LOS availability from adjacent peek positions
SQUAD_DISPERSION_WEIGHT	0.12	Spread out from squadmates
DISTANCE_WEIGHT	0.10	Proximity to search center
FORMATION_POSITION_WEIGHT	0.08	Match formation ideal position
FIGHTABILITY_BONUS (additive)	+0.25 (HALF) / +0.15 (FULL)	Added if <code>canShootFrom()</code>
blindPenalty (subtractive)	-0.50	Full cover with no peek angle to target

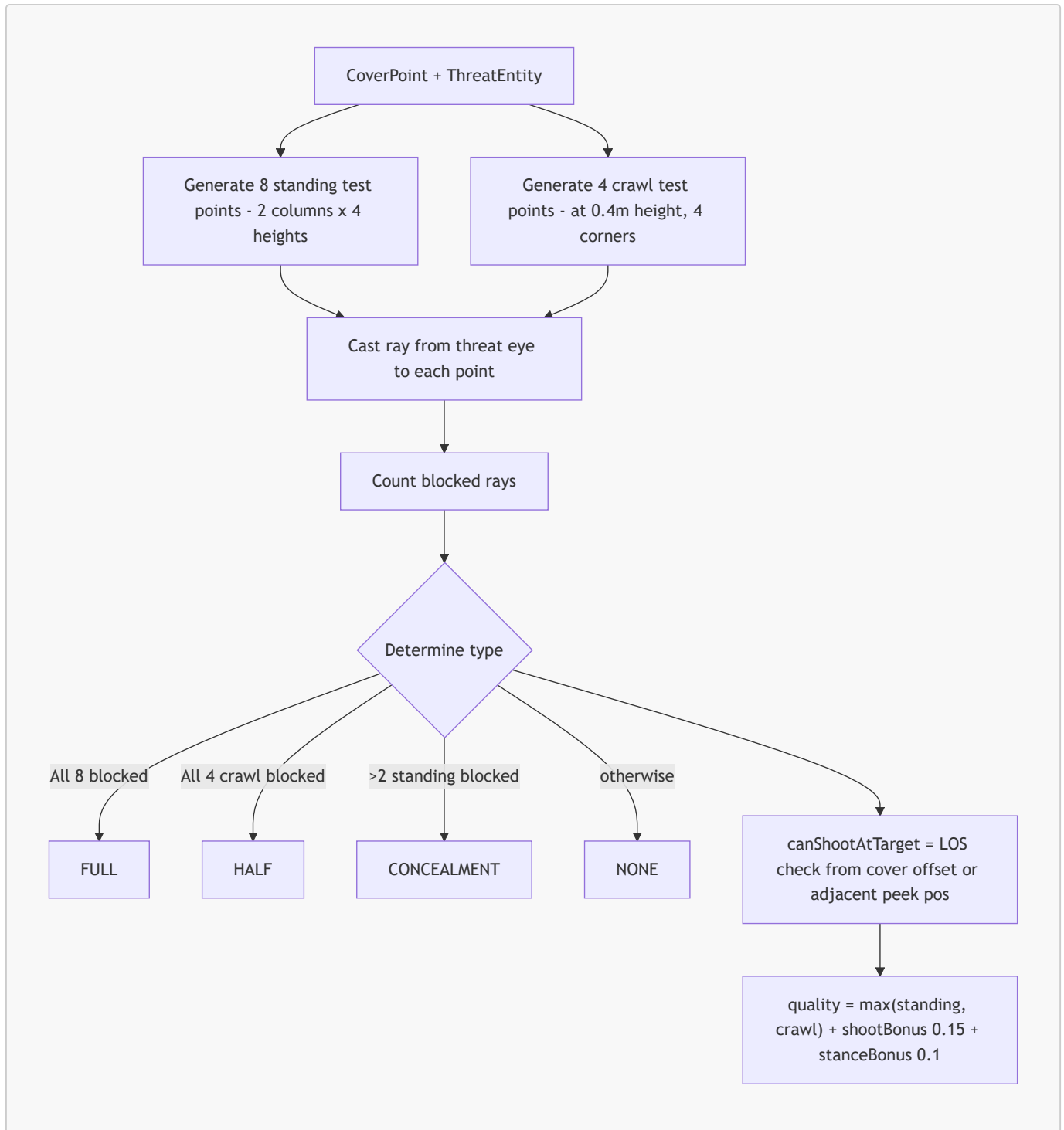
2.3 Cover Height Classification



Blocks excluded from valid cover: `IronBarsBlock`, `GlassBlock`, `StainedGlassBlock`, `TintedGlassBlock`.

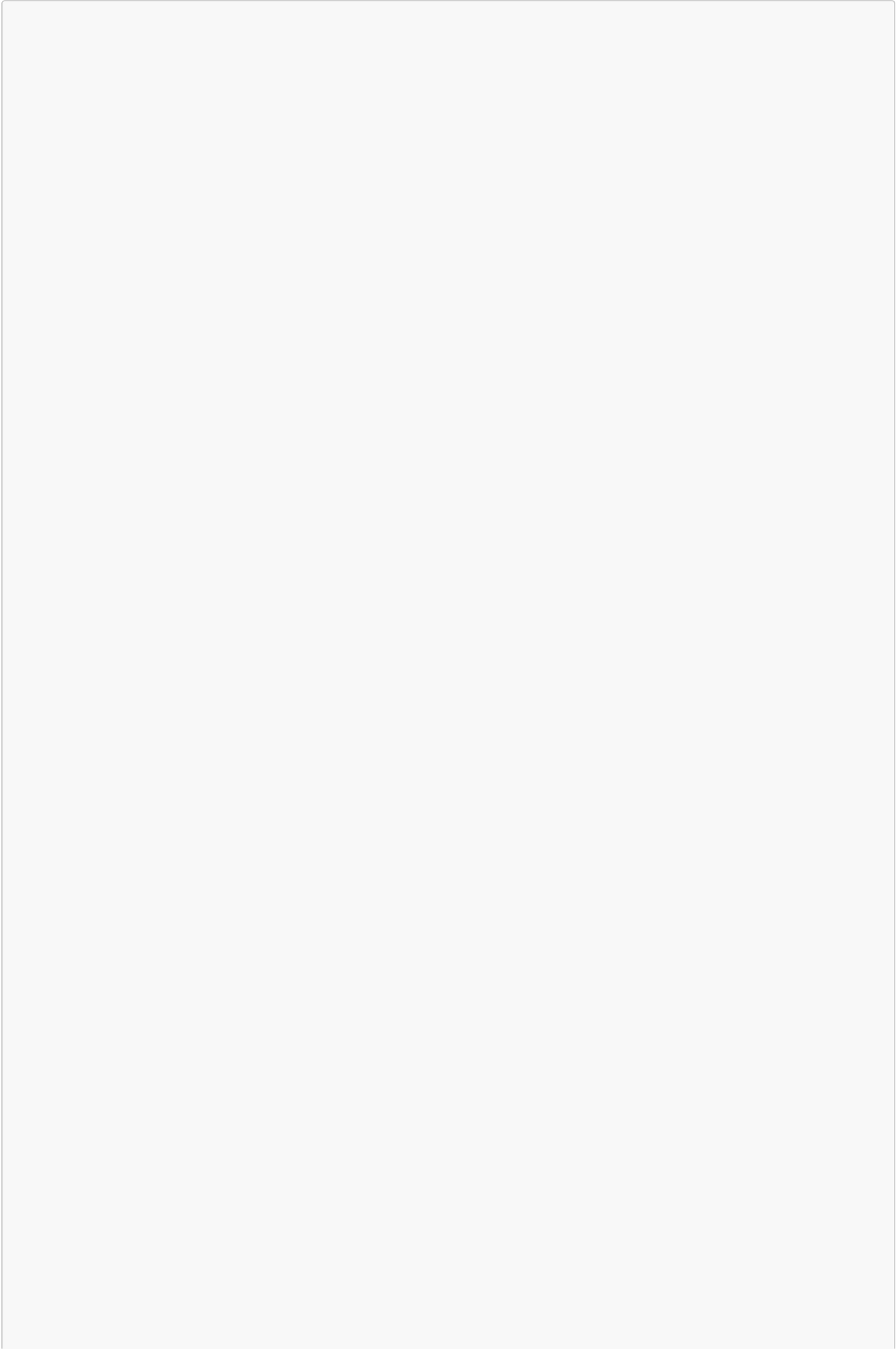
2.4 Cover Quality Evaluator (Raycast)

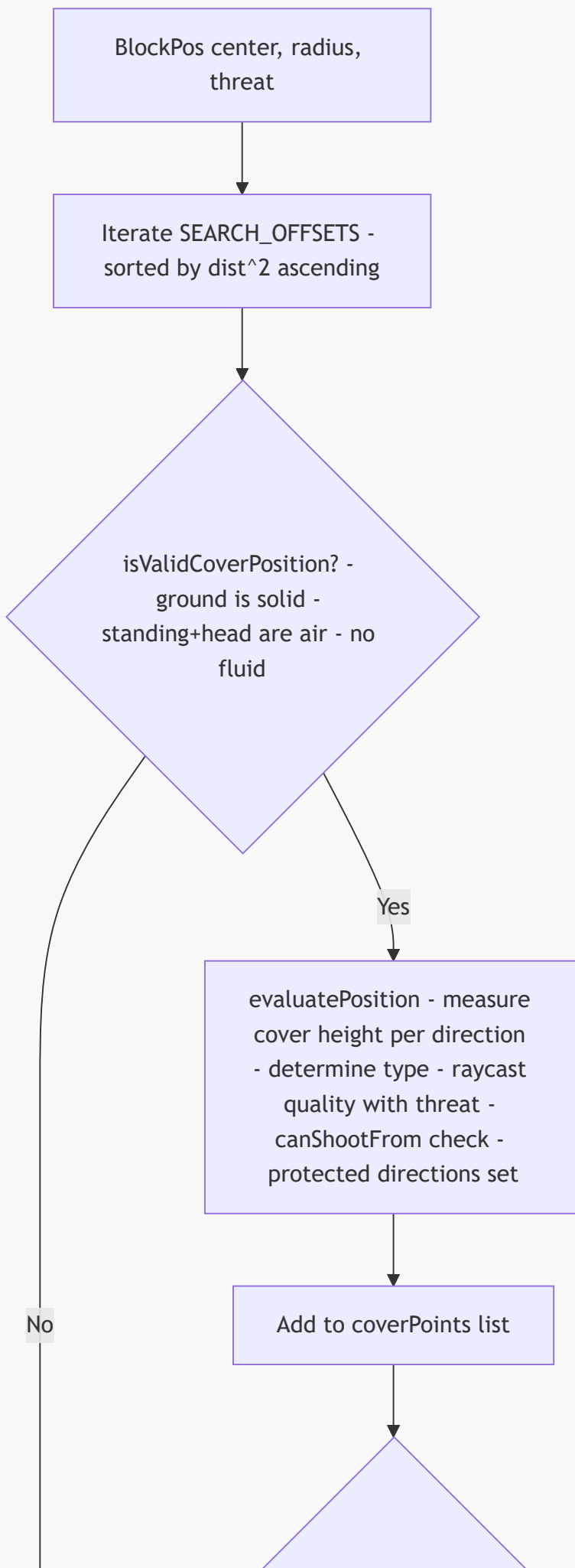
`CoverQualityEvaluator` casts 8 standing-height rays + 4 crawl-height rays from the cover position toward the threat's eye position:

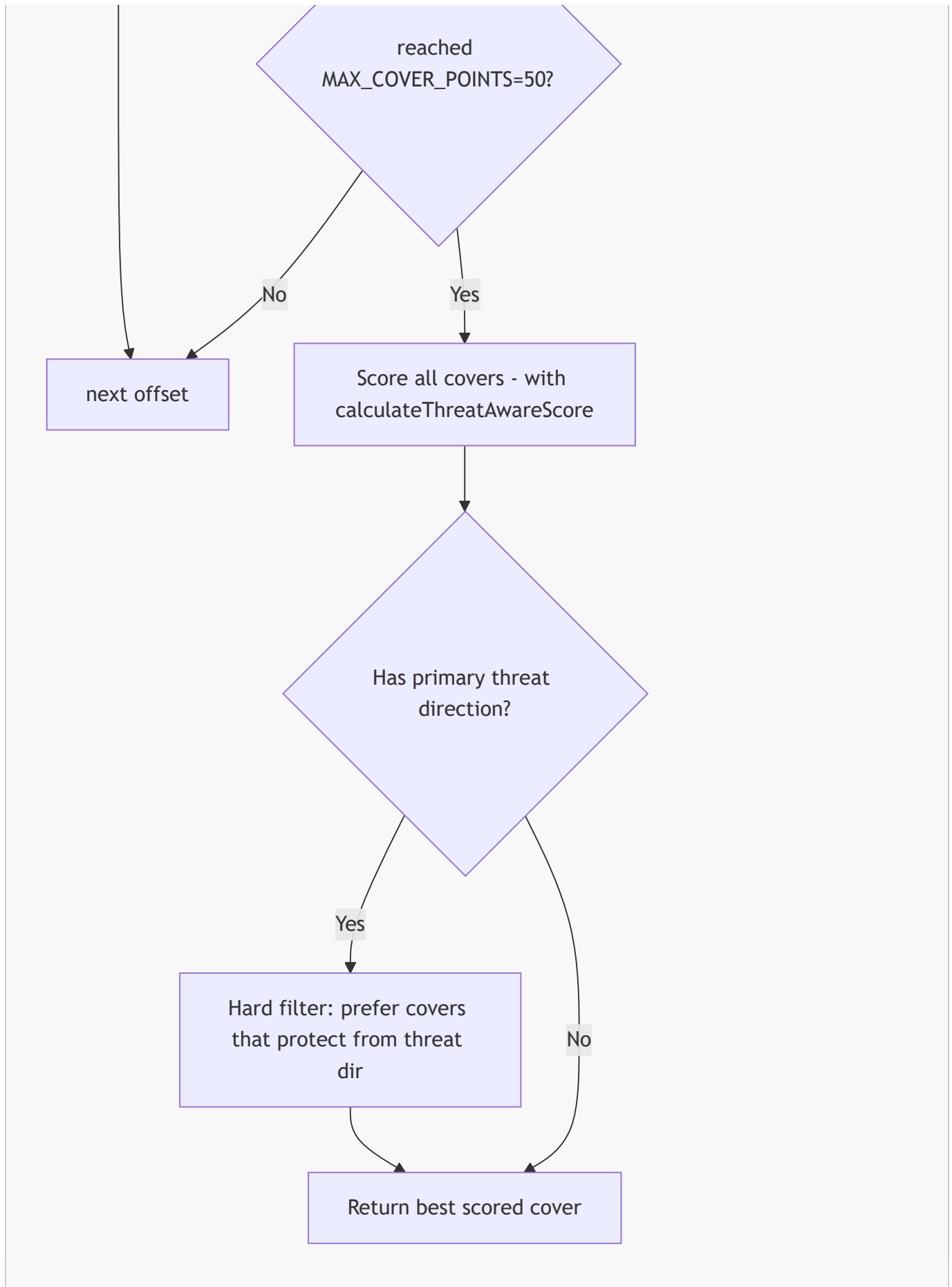


2.5 Cover Finder

Search: Inside-out spiral search over **SEARCH_OFFSETS** (pre-calculated, sorted by distance squared). Radius = 12 blocks, max 50 cover points.







2.6 Cover Reservation Manager

- Static `ConcurrentHashMap<BlockPos, Set<UUID>>` -- thread-safe reservations
- `MAX_RESERVATIONS_PER_COVER = 1` (one soldier per cover)

- `RESERVATION_TIMEOUT_MS = 30000` (auto-cleanup)
- Released on `clearCover()` or `clearTargetCover()`
- Blacklist system in `CoverTacticalGoal` with reasons: `PATH_FAILED`, `STUCK_SEEKING`, `STUCK_REPOSITIONING`
- Blacklist cleared every 15s

2.7 Peek Count Penalty

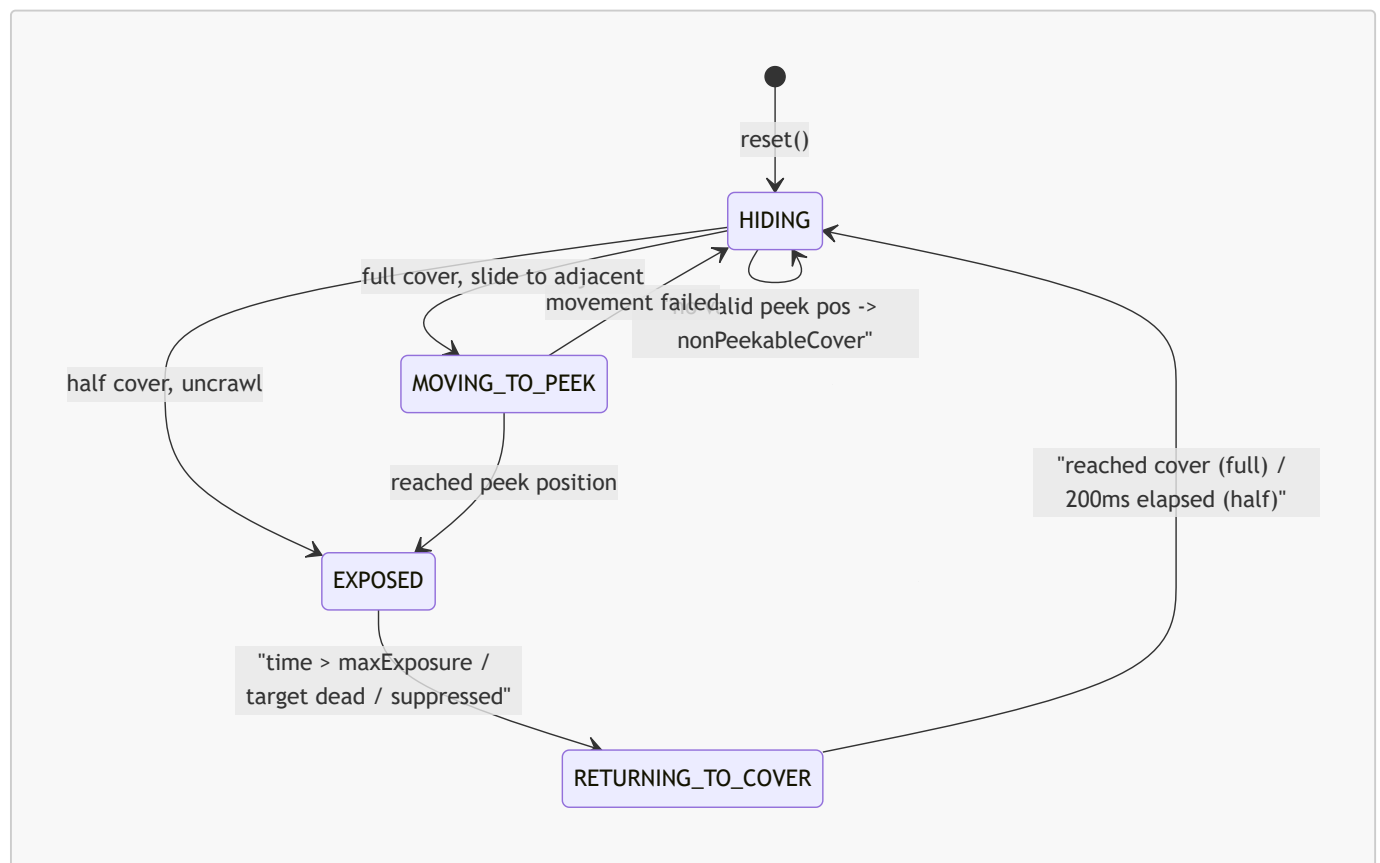
After 4 peeks from the same cover, a quality penalty is applied:

```
extraPeeks = peekCount - PEEK_COUNT_PENALTY_THRESHOLD + 1 // (4)
penalty = min(MAX_COVER_PENALTY(0.60), extraPeeks * 0.15)
```

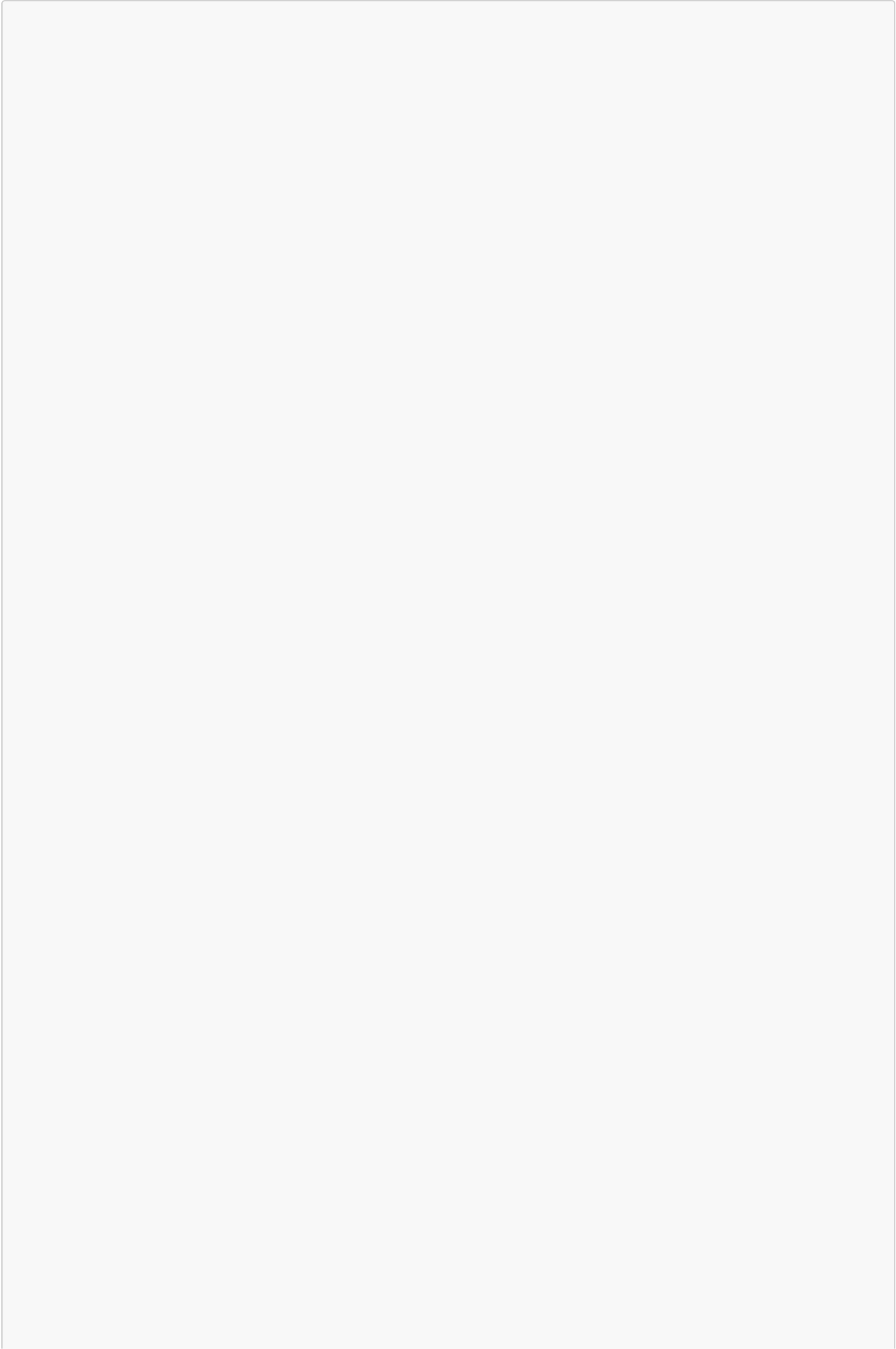
This penalty is fed back into the hysteresis check when evaluating better cover.

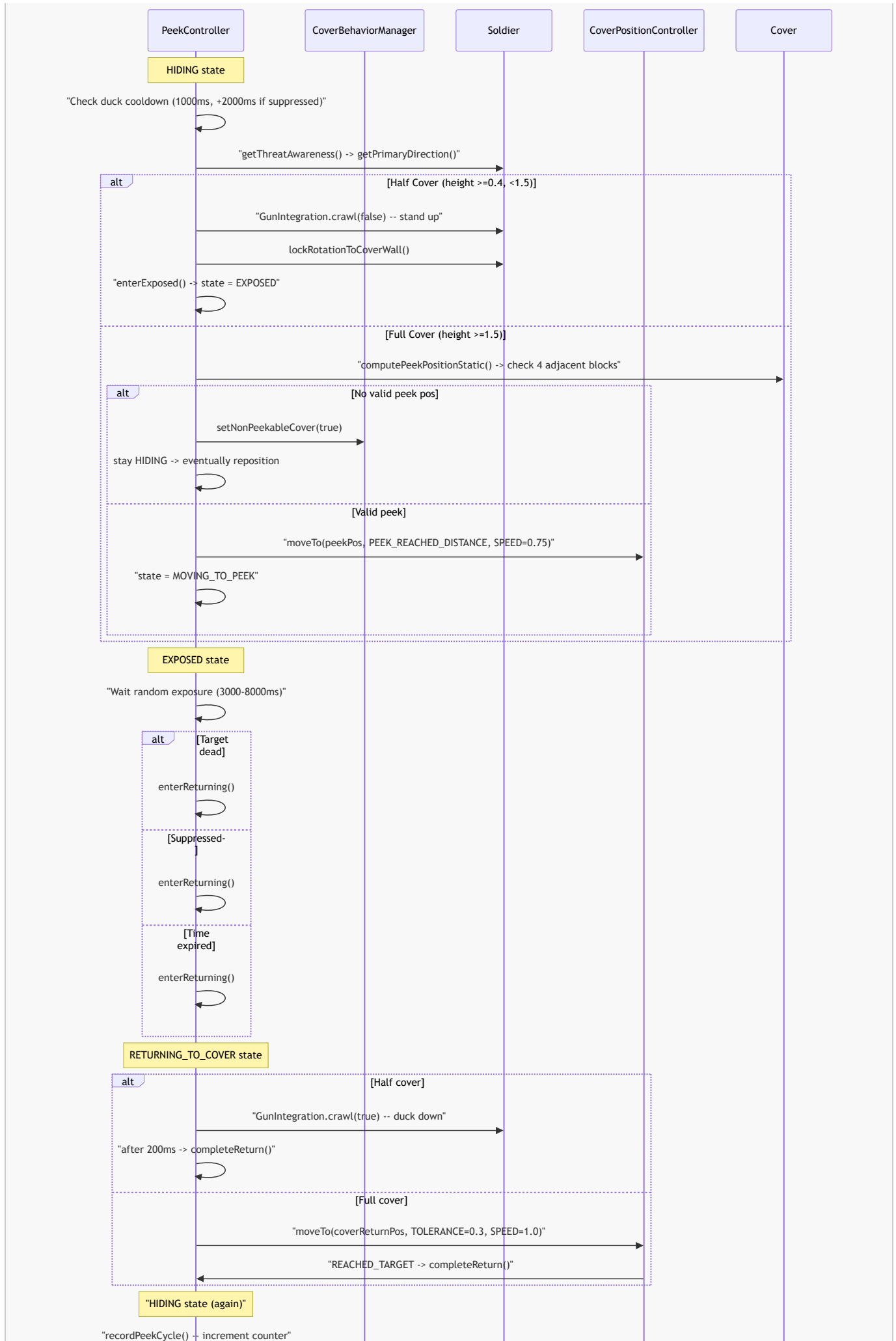
3. Peek System

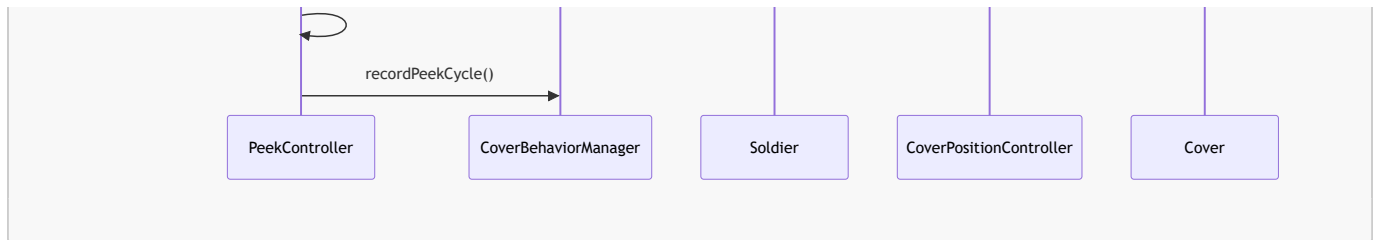
File: `entity/ai/PeekController.java`



3-state Peek Flow







Timing Constants

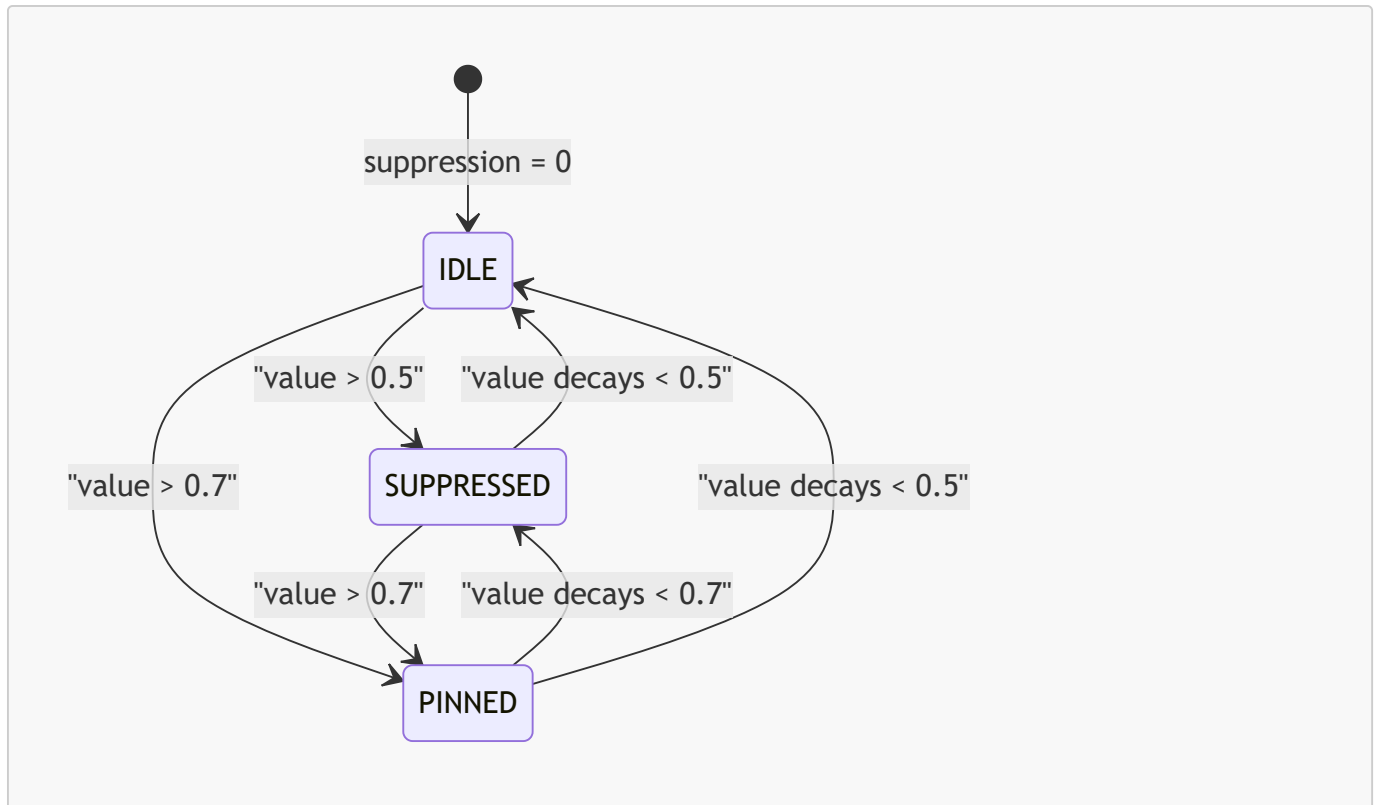
Constant	Value	Purpose
EXPOSURE_TIME_MIN_MS	3000	Minimum exposure
EXPOSURE_TIME_MAX_MS	8000	Maximum exposure
DUCK_COOLDOWN_MS	1000	Min time between peeks
SUPPRESSED_HIDE_EXTRA_MS	2000	Extra hide when suppressed
PEEK_REACHED_DISTANCE	0.05	Distance threshold for "reached"
PEEK_SPEED	0.75	Velocity slide speed to peek
RETURN_SPEED	1.0	Velocity slide speed returning
NON_PEEKABLE_REPOSITION_TICKS	60	Ticks before reposition (~3s)

Full Cover Peek Mechanics

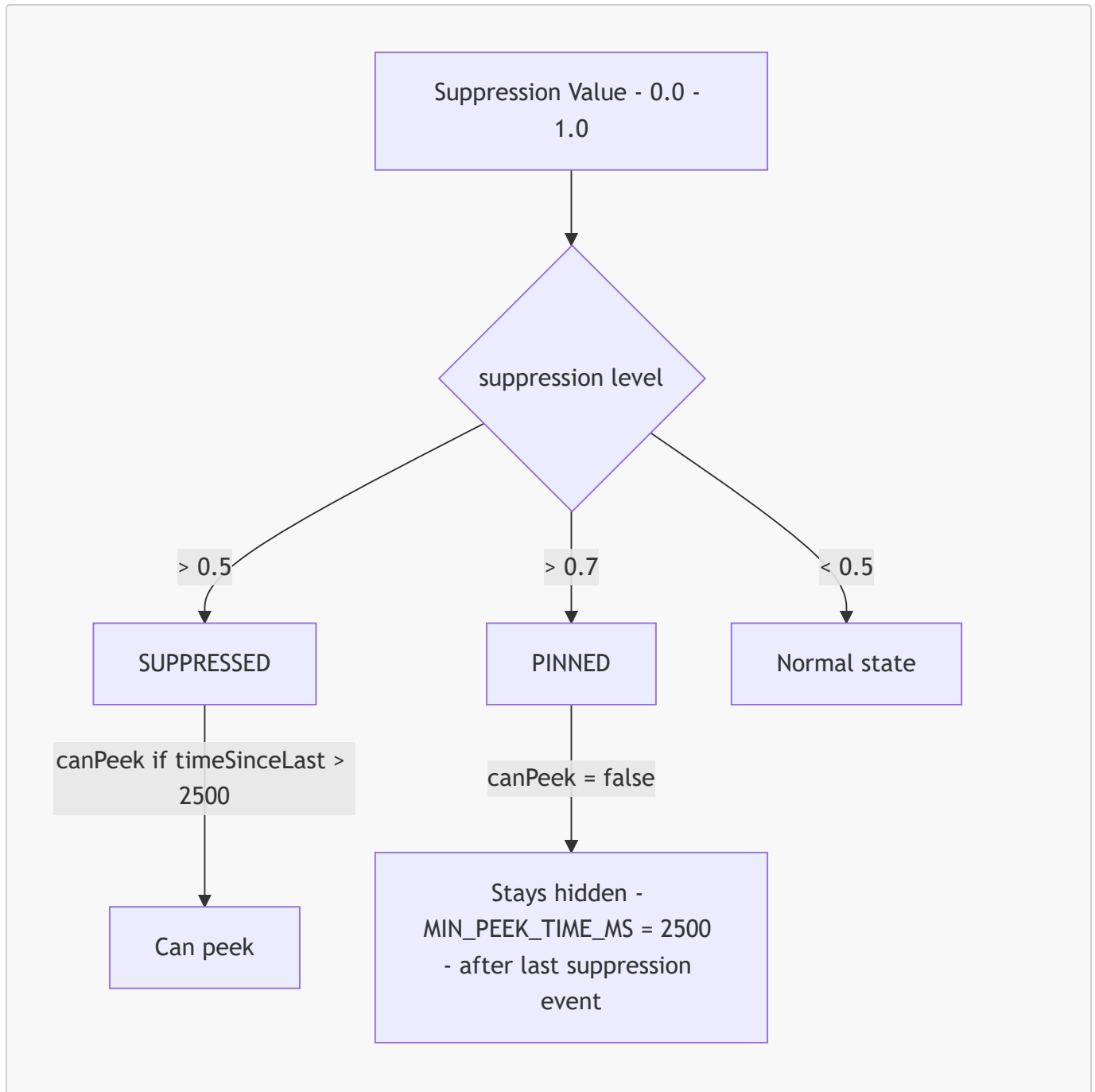
- Soldier velocity-slides 1 block sideways to an adjacent standable position
- Position is validated: ground solid, standing+head air, not fluid
- LOS is validated from the peek position (direct ray to target, then cone fallback)
- On return, slides back to original cover position
- Half-cover: instant state change, soldier uncrawls/crawls

4. Suppression System

4.1 SuppressionTracker ([combat/cover/SuppressionTracker.java](#))

**Decay:**

- `DECAY_RATE` = 0.15 base per tick
- `decayMultiplier` = 2.0 when in cover, 1.0 when not
- Applied as: `decay` = `0.15` x `decayMultiplier` x `0.05` = `0.0075/tick` (in cover)
- `peakSlowdown`: higher peak suppression = slower decay (1.0 peak -> 50% decay speed)



Suppression sources:

Source	Base Value	Multipliers
Near miss (❤ blocks)	0.25	distanceFactor x speedMultiplier x burstMultiplier
Direct incoming fire	0.30	speedMultiplier
Damage taken	0.50	--

Burst suppression decay: $\text{burstMultiplier} = 1.0 - \min(\text{burstCount} - 1, 3) \times 0.15$

- First hit: 1.0x, Second hit: 0.85x, Third: 0.70x, Fourth+: 0.55x

4.2 IncomingFireHandler (`combat/cover/IncomingFireHandler.java`)

- `@SubscribeEvent onLivingHurt`: calls `coverManager.onTakeDamage()` and `coverManager.onIncomingFire()` for all `SoldierEntity` damage

- `@SubscribeEvent onEntityJoin`: tracks `EntityKineticBullet` entities
- `tick()`: for each tracked bullet, computes line segment between previous and current position, then `checkNearMissLineSegment()` against all nearby soldiers within `NEAR_MISS_THRESHOLD` (3 blocks)
- `@SubscribeEvent onLivingDeath`: propagates death to `SquadThreatIntel.markThreatDead()` and each squad member's `SoldierCombatGoal.onTargetKilledByTeammate()`

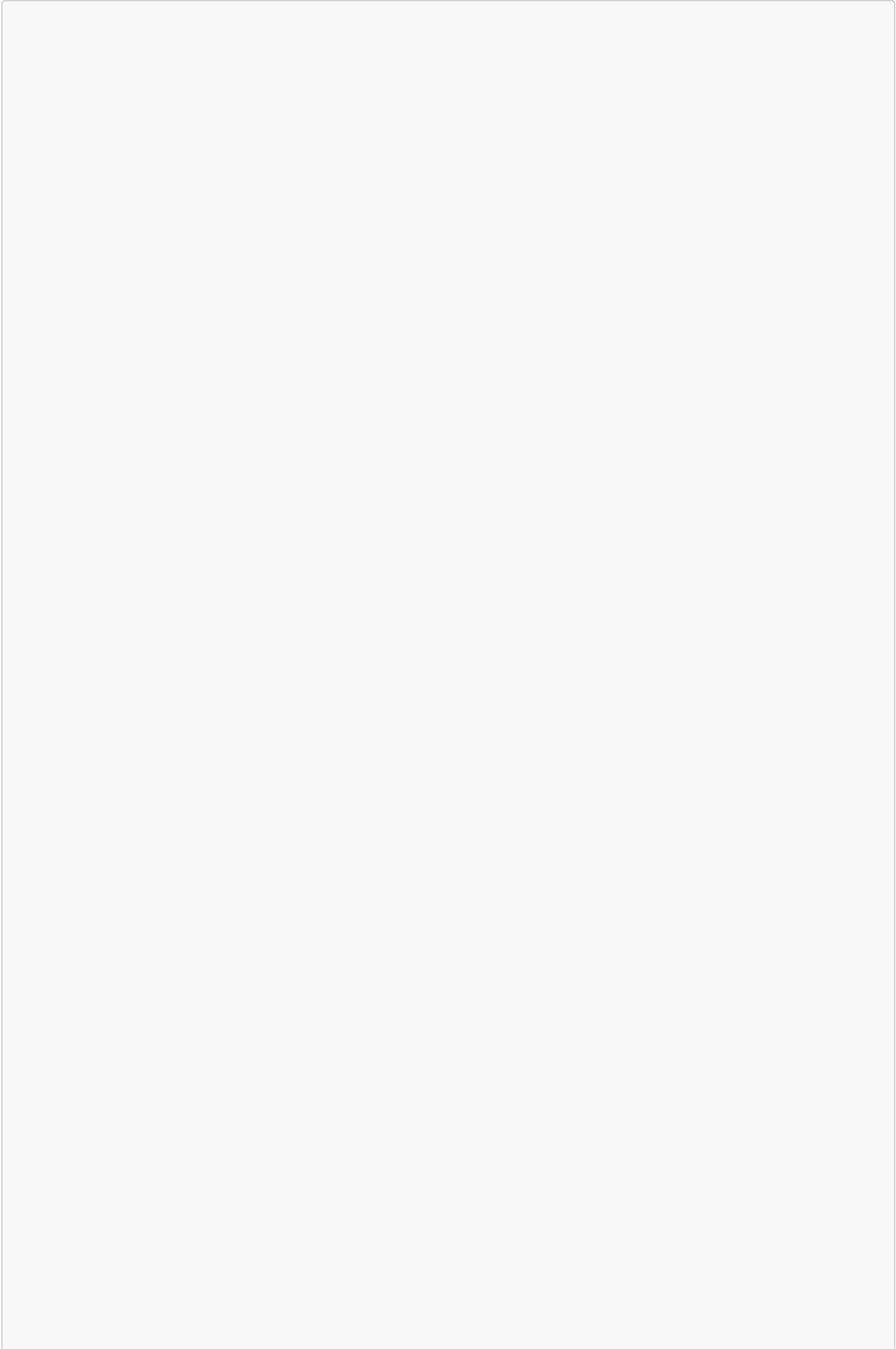
4.3 Suppression Fire Integration

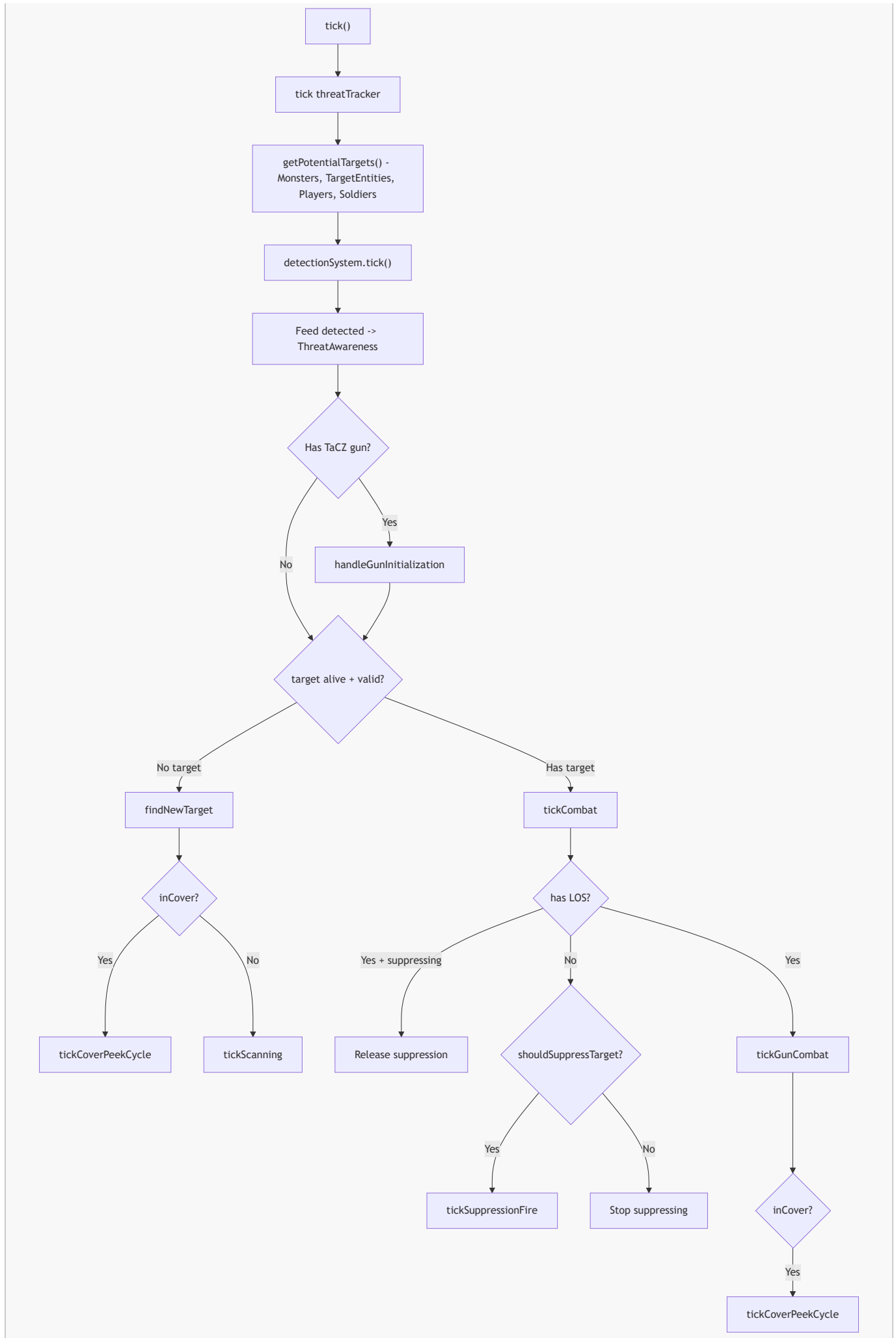
See [Section 15: Suppression Fire Lifecycle](#).

5. Soldier Combat Goal

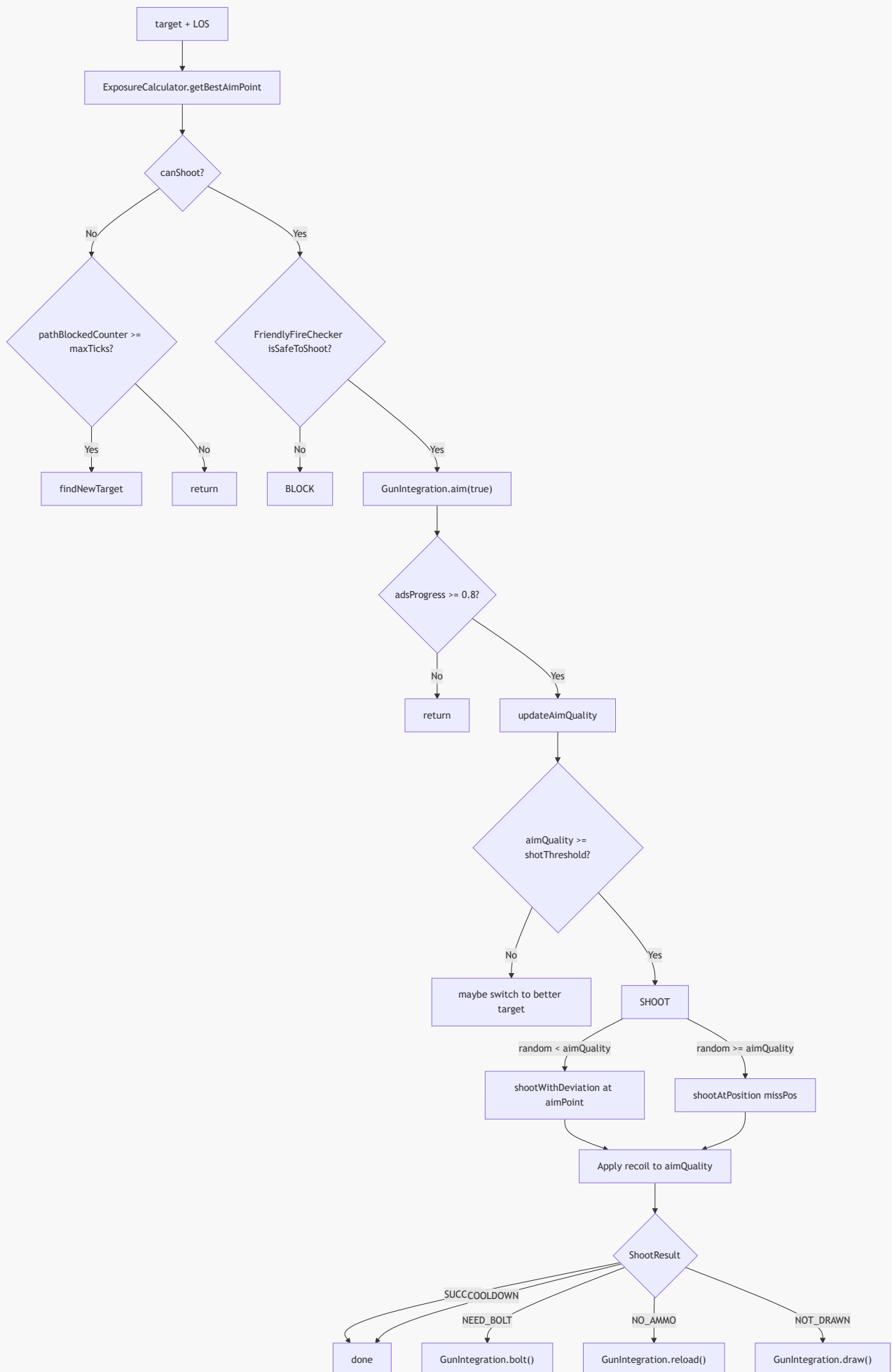
File: `entity/ai/SoldierCombatGoal.java`

Main Tick Flow





Shooting Mechanics (**tickGunCombat**)



Target Acquisition Priority (`findNewTargetInternal`)

1. **Forced target** (from SEND ping) -> nearest entity within 20 blocks with LOS
2. **Ping threat** (ENEMY ping) -> nearest entity within 20 blocks with LOS
3. **Any LOS target** -> highest hit probability
4. **Threat-direction fallback** -> entity closest to primary threat direction
5. **Last known position** -> entity nearest to last known position (20 block radius)
6. **Closest entity** -> nearest entity regardless of LOS

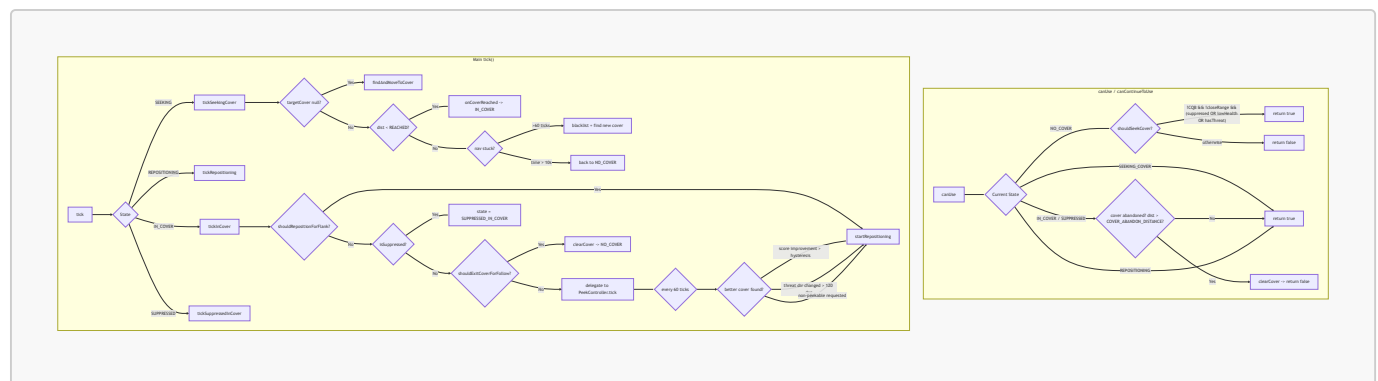
Aim Quality

- Builds toward `targetAimQuality` via lerp with `buildRate`
- Decays when moving (own movement), when target moves, or when LOS is lost
- Shot threshold = `targetAimQuality` x `thresholdScale` (min 0.15)
- Slower guns (manual bolt) use a higher `thresholdScale`

6. Cover Tactical Goal

File: `entity/ai/CoverTacticalGoal.java`

Full State Machine + Decision Flow



Cover Search Center Selection (`findAndMoveToCover`)



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Path Navigation to Cover (`moveToCover`)



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Hysteresis for Cover Switching

When evaluating better cover while IN_COVER:

```
hysteresisFactor = penalty > 0 ? 1.0 : 1.0 + HYSTERESIS_THRESHOLD(0.35)
currentScore = currentCover.quality x hysteresisFactor - penalty
newScore = newCover.quality
switch if: newScore > currentScore
```

This means a new cover must be at least 35% better to justify switching, unless the current cover has a peek-count penalty.

7. Squad System

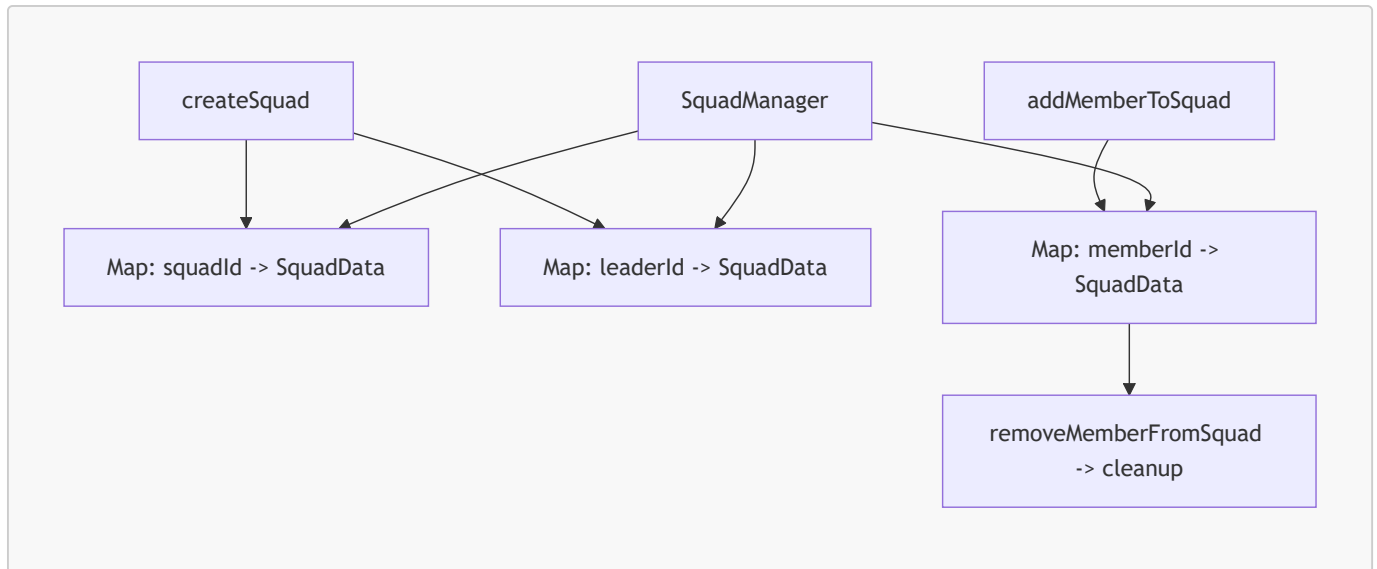
Package: `com.stevesarmy.squad`

7.1 SquadData

```
SquadData {
    UUID squadId          -- unique squad identifier
    UUID leaderId         -- owner UUID or 0000...0001 for enemies
    List<UUID> memberIds  -- soldier UUIDs
    SquadMode mode        -- FOLLOW or HOLD
    SquadFormation formation -- NONE, LINE, WEDGE, COLUMN, DIAMOND, CQB
    boolean cqbMode       -- close-quarters battle mode flag
    SquadThreatIntel threatIntel -- shared threat knowledge
}
```

7.2 SquadManager (SavedData)

- Stored as `steves_army_squads` in dimension data
- Three maps: `squadsByLeader`, `squadsByMember`, `squadsById`
- Key methods: `createSquad`, `addMemberToSquad`, `removeMemberFromSquad`, `disbandSquad`, `setSquadFormation`, `setSquadCQB`



7.3 Squad Cover Context

`SquadCoverContext` record:

```

record SquadCoverContext(
    boolean inSquad,
    SquadFormation formation,
    int squadSize,
    int memberIndex,
    List<BlockPos> occupiedCovers,
    List<Vec3> squadThreatDirections
)
  
```

Used by `CoverFinder` to:

- Apply `SQUAD_DISPERSION_WEIGHT` (0.12): penalize covers too close (<4 blocks) to squadmates
- Apply `FORMATION_POSITION_WEIGHT` (0.08): reward covers near the ideal formation offset

7.4 SquadFormation

Formation	Color	Use
NONE	White	Default, no formation effect
LINE	Green	Soldiers spread horizontally
WEDGE	Blue	V-shaped, forward-oriented
COLUMN	Orange	Single-file line
DIAMOND	Magenta	Diamond pattern
CQB	Red	Close-quarters, overrides cover seeking

7.5 Squad Modes

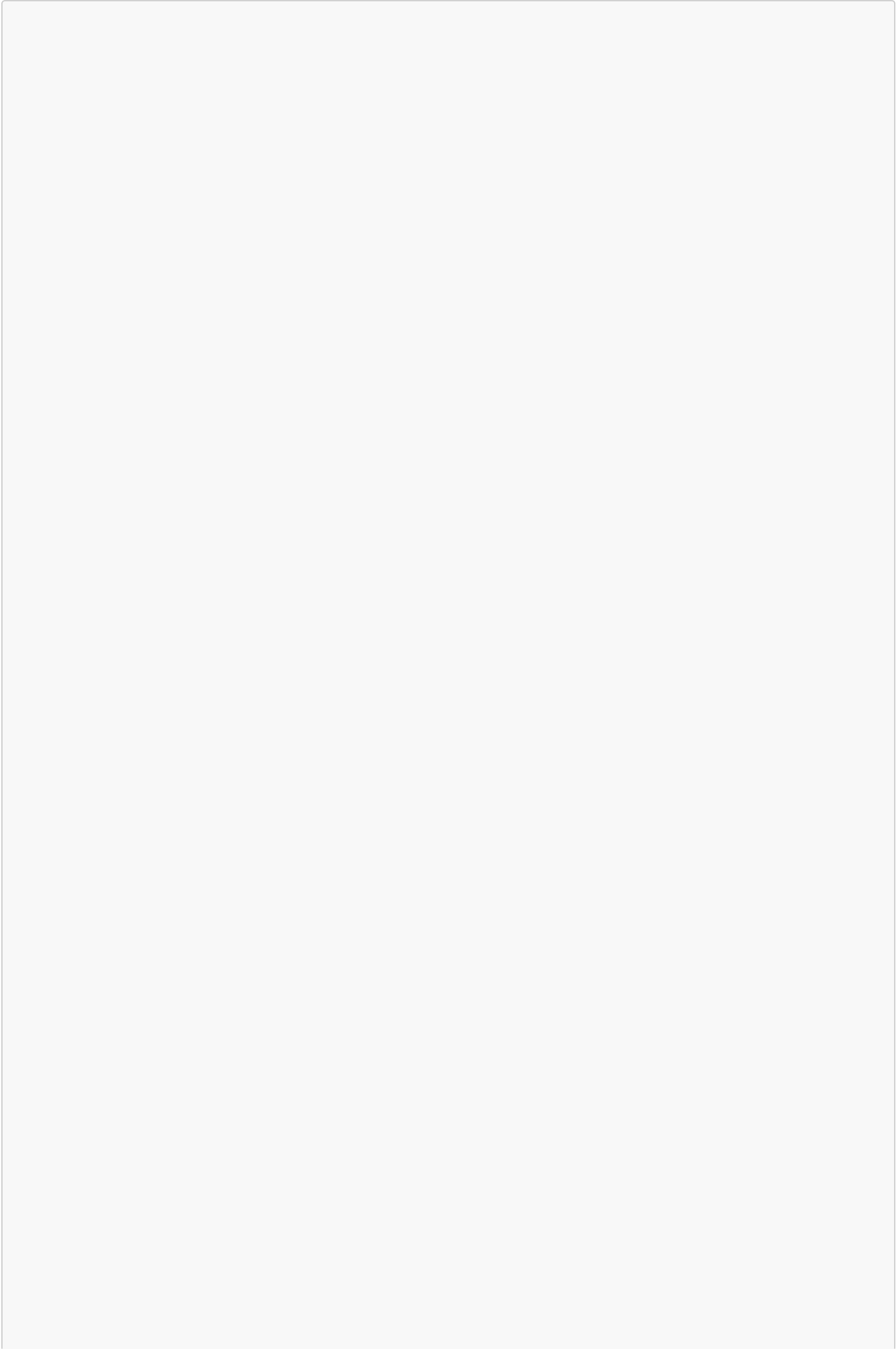
```
SquadMode.FOLLOW -- soldier follows owner, seeks cover near owner  
SquadMode.HOLD   -- soldier holds position, seeks cover near hold pos
```

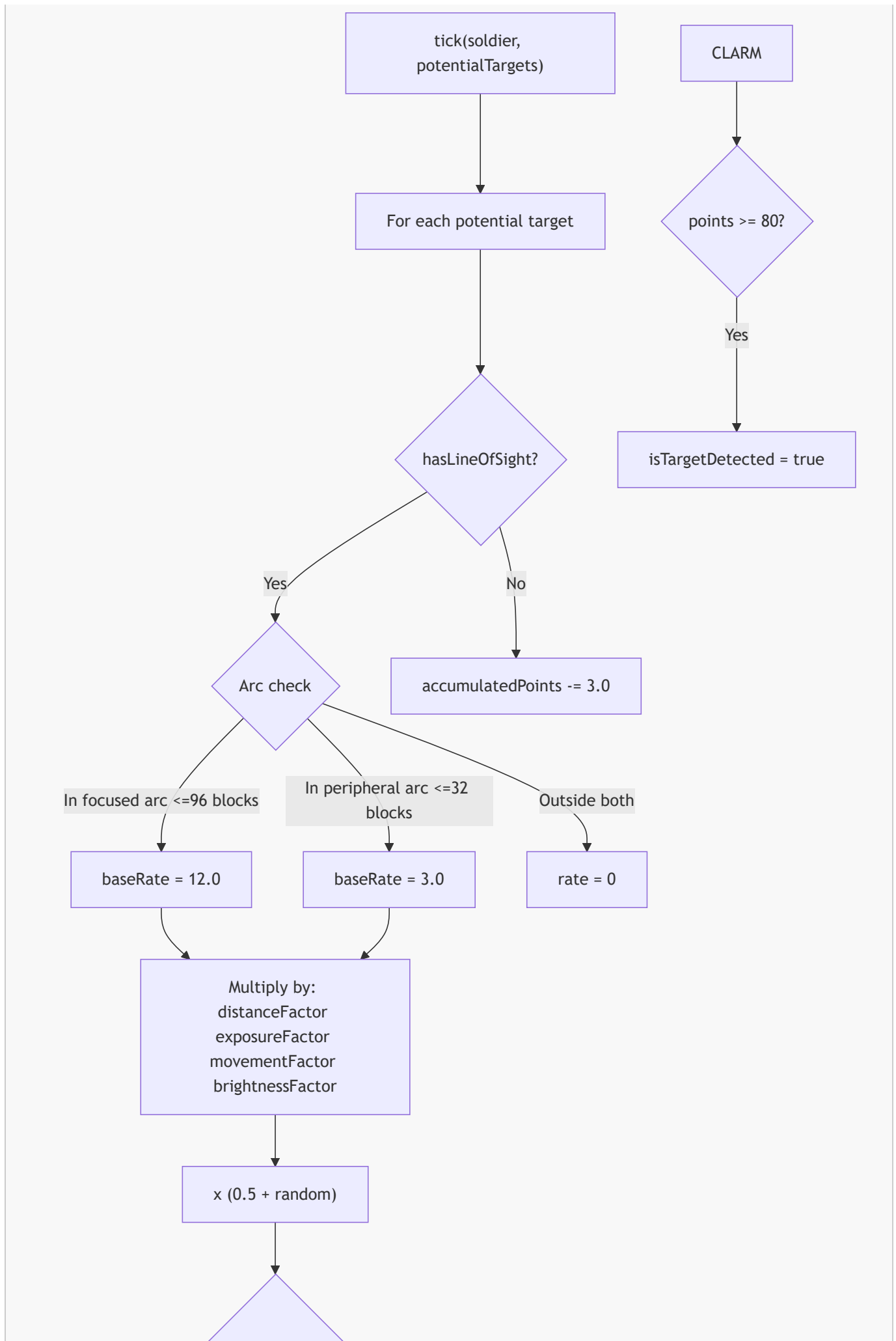
7.6 SquadThreatIntel

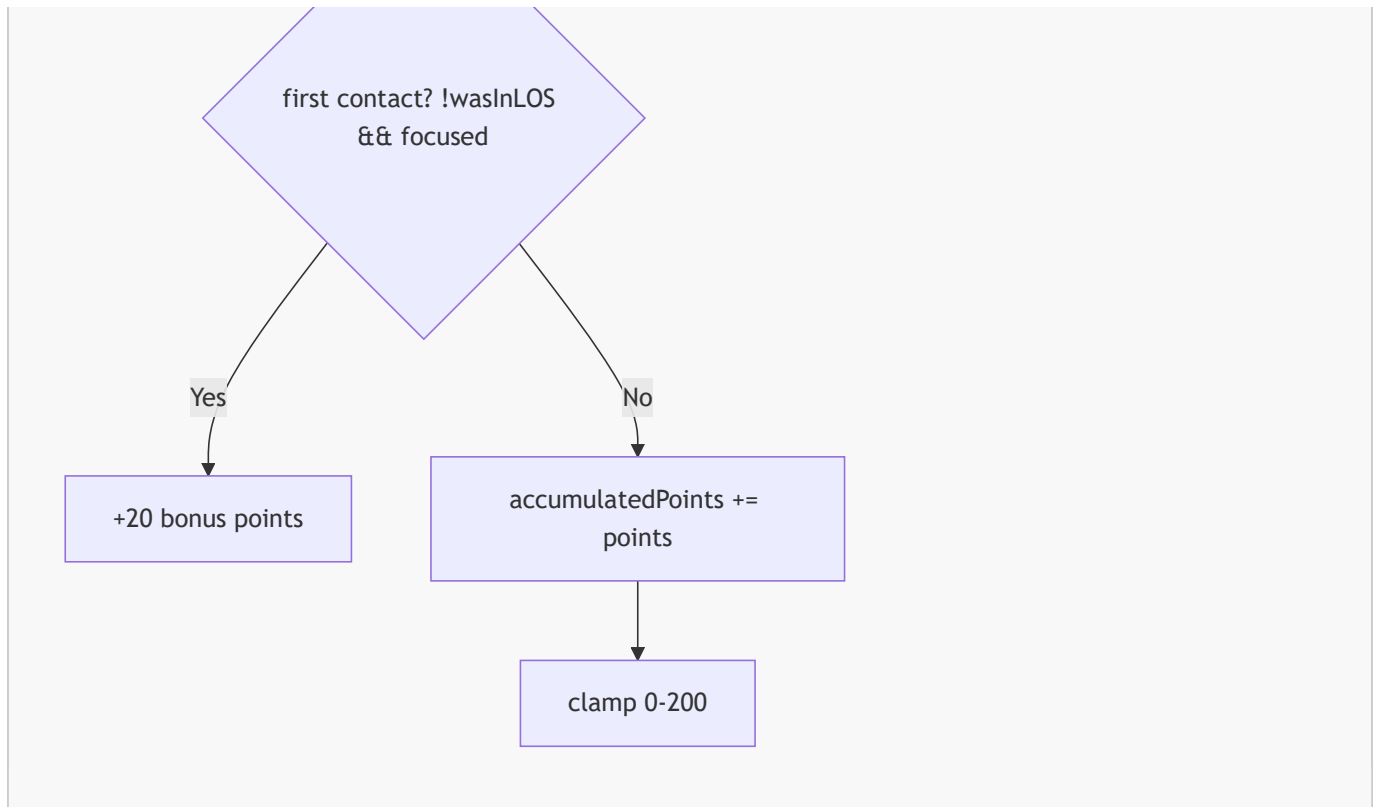
- Shared threat knowledge for the squad (stored in `SquadData`)
 - Each `ThreatKnowledge` tracks: position, last seen time, reporter, accuracy, alive/dead, suppression state
 - Threats remembered for `THREAT_MEMORY_TICKS = 600` (30s)
 - Stale check: `STALE_TIMEOUT_TICKS = 60` (3s) for suppression assignment eligibility
 - Suppression heartbeat timeout: 10 ticks -- if suppressor doesn't heartbeat, assignment is cleared
-

8. Detection System

File: `combat/DetectionSystem.java`







Detection Arcs

Arc	Range	Rate	Angle
Focused	96 blocks	12.0 pts/tick	+/-45 deg from look direction
Peripheral	32 blocks	3.0 pts/tick	+/-90 deg from look direction (excl. focused)

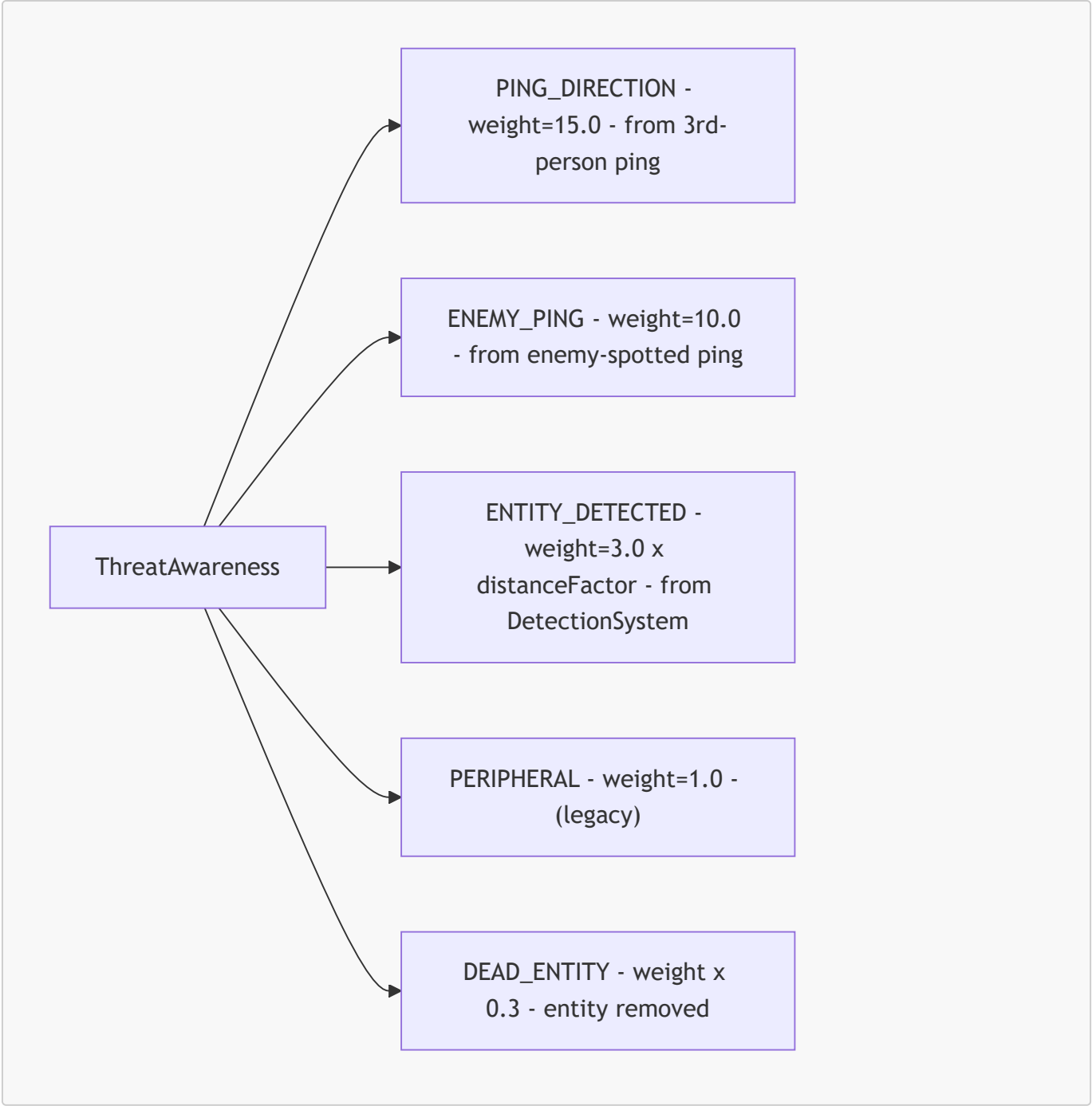
Detection Factors

Factor	Range	Notes
Distance	0.0-1.0+	1.0 at 16 blocks, drops quadratically beyond
Exposure	0.0-1.0	From <code>ExposureCalculator</code> (% of body visible)
Movement	0.3-1.5	Sprint=1.5, walk=1.0, sneak=0.3, still=0.7
Brightness	0.3-1.0	$0.3 + 0.7 \times \text{sqrt}(\text{lightLevel} / 15)$

9. Threat Awareness

File: `combat/ThreatAwareness.java`

Threat Sources



Smooth Direction

- Updated via `updateSmoothDirection(threatPos)` with configurable `blendFactor`
- Decays after `threatSmoothDecayTimeMs` ms of no updates
- Used by `CoverTacticalGoal` for threat direction and peek direction

Query Methods

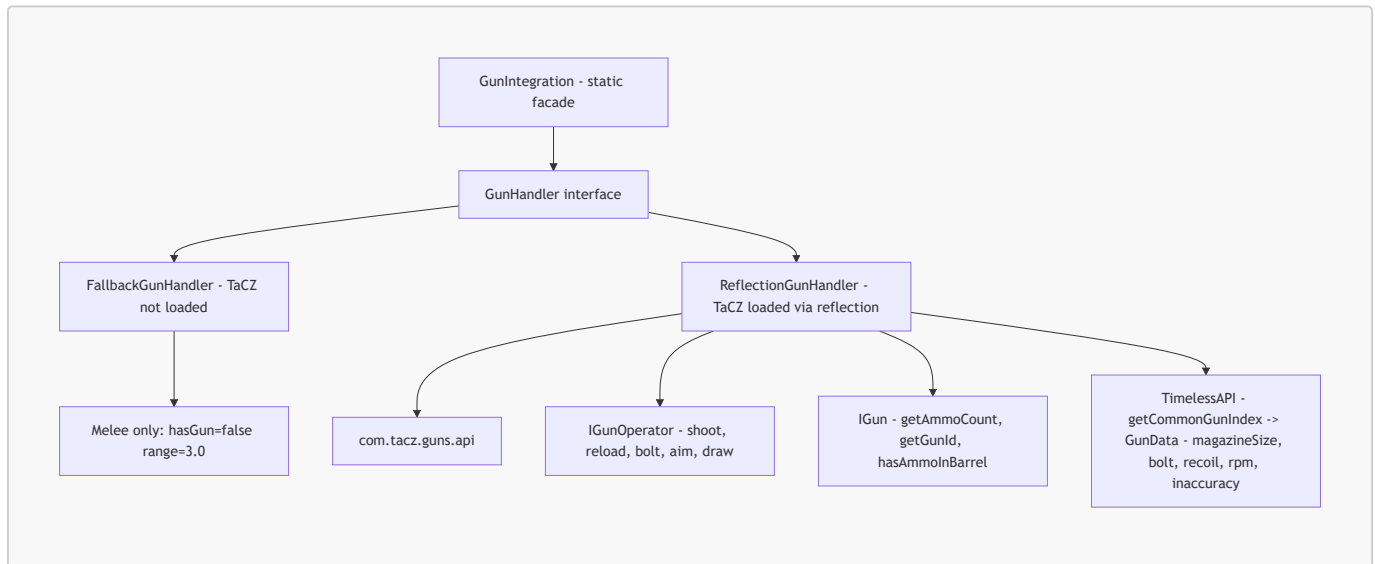
Method	Returns	Use Case
<code>getPrimaryDirection(pos)</code>	Normalized Vec3	Cover protection scoring
<code>getThreatDirectionForProactivePeek(pos)</code>	Vec3 or null	PeekController HIDING check
<code>getWeightedAveragePosition()</code>	Vec3	Debug rendering, fallback targeting
<code>isBeingFlanked(pos)</code>	Boolean	Flank-driven repositioning

Method	Returns	Use Case
<code>getThreatCoverage(coverPos)</code>	0.0-1.0	Threat protection assessment

10. Gun Integration

File: `combat/GunIntegration.java`

Architecture



Gun Operations (ReflectionGunHandler)

Operation	Reflection Target	Notes
<code>hasGun(entity)</code>	<code>IGun.mainHandHoldGun</code>	--
<code>shoot(shooter, target)</code>	<code>IGunOperator.shoot(pitch, yaw)</code>	Aim at entity center
<code>shootWithDeviation(shooter, aimPoint, ...)</code>	<code>IGunOperator.shoot(pitch, yaw)</code>	Uses <code>ExposureCalculator.AimPointResult</code> , with pitch/yaw deviation
<code>shootAtPosition(shooter, pos)</code>	<code>IGunOperator.shoot(pitch, yaw)</code>	Aim at arbitrary Vec3
<code>reload(entity)</code>	<code>IGunOperator.reload</code>	Checks ammo, logs
<code>bolt(entity)</code>	<code>IGunOperator.bolt</code>	--
<code>aim(entity, isAiming)</code>	<code>IGunOperator.aim(boolean)</code>	--
<code>crawl(entity, isCrawl)</code>	<code>IGunOperator.crawl</code>	Also sets soldier <code>setCrawling(boolean) -> Pose.SWIMMING</code>
<code>initialData(entity)</code>	<code>IGunOperator.initialData</code>	Initialize TaCZ gun data
<code>draw(entity)</code>	<code>IGunOperator.draw(ItemStack)</code>	Draw gun (skipped if reloading)

ShootResult Mapping

TaCZ Enum	Mapped Result
SUCCESS	SUCCESS
NO_AMMO	NO_AMMO
COOL_DOWN	COOLDOWN
NEED_BOLT	NEED_BOLT
IS_BOLTING	IS_BOLTING
IS_RELOADING	IS_RELOADING
IS_DRAWING	IS_DRAWING
NOT_DRAW	NOT_DRAWN
NOT_GUN	NOT_GUN

Ammo Management

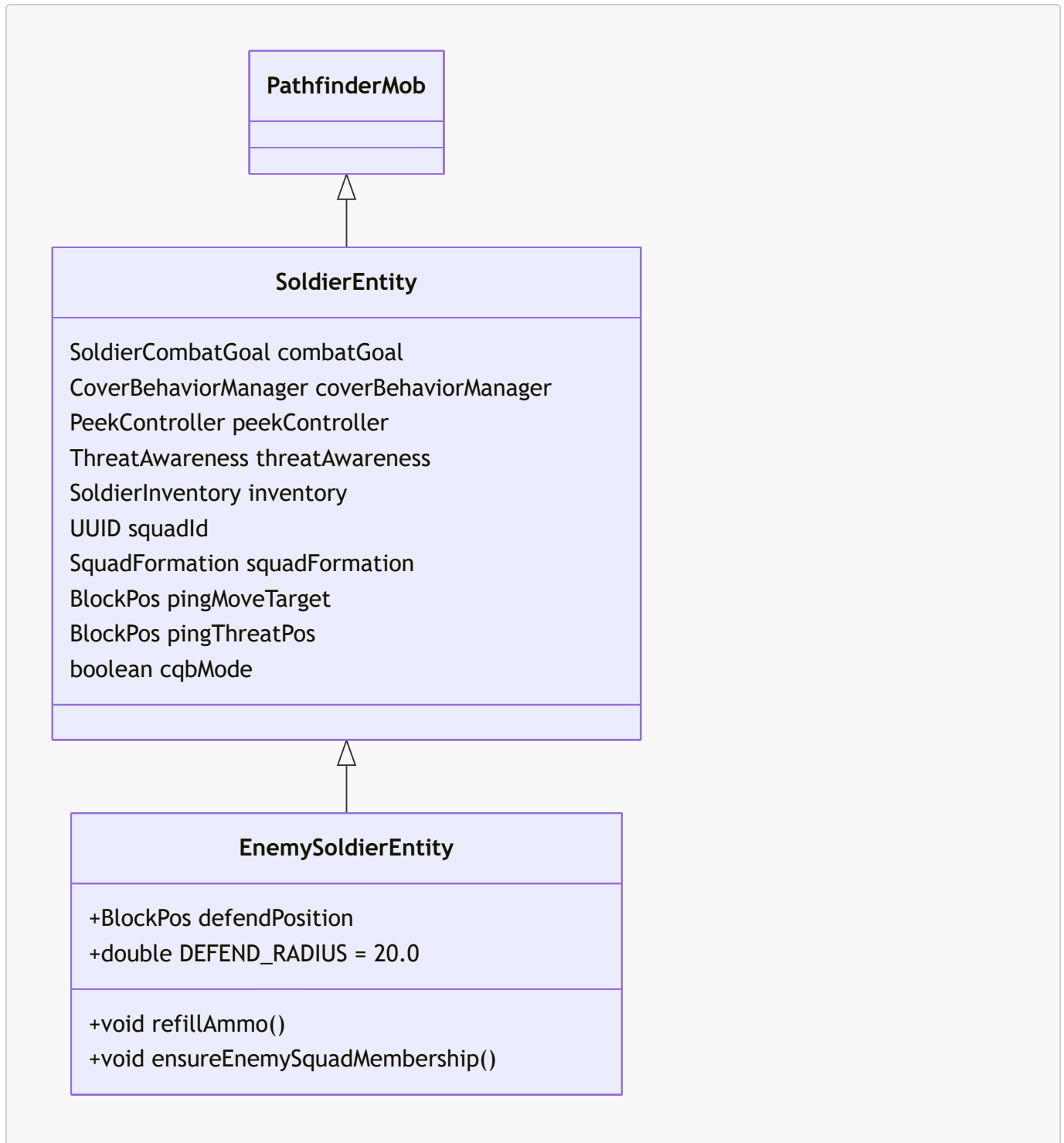
- Ammo is consumed from the 9-slot soldier inventory
- `useInventoryAmmo()` checks TaCZ gun data config
- When magazine is empty, `SoldierCombatGoal` auto-reloads
- `EnemySoldierEntity.refillAmmo()` refills to full every tick (infinite ammo for enemies)

Accuracy Values (InaccuracyType)

```
STAND = 5.0
MOVE  = 5.75
SNEAK = 3.5
LIE   = 2.5
AIM   = 0.15
```

Soldiers aim before shooting (ADS, `aimProgress` $\geq$ 0.8 or 0.5 for suppression).

11. Entity Hierarchy



SoldierEntity Key Features

- **Move control:** `CoverPositionController` (custom, handles velocity-based peek sliding)
- **Navigation:** `SoldierGroundNavigation` (custom ground nav)
- **Inventory:** 9-slot `SoldierInventory` (1 main hand + 1 armor slots + 7 general)
- **Data syncing:** 18+ `EntityDataAccessor` fields for cover state, peek state, suppression, threat direction, debug data -- all synced from server to client for HUD rendering
- **Friendly fire check:** `isFriendlyTo()` -- same owner = friendly

EnemySoldierEntity Extras

- No `SoldierMoveToPingGoal` (no ping guidance)
- No follow/hold goals -- uses `EnemyDefendGoal` to stay near spawn

- Auto-joins enemy squad (leader UUID = 00000000-0000-0000-0000-000000000001)
 - Auto-refills ammo every tick (infinite)
 - `canAttack()` returns false for other `EnemySoldierEntity` instances
-

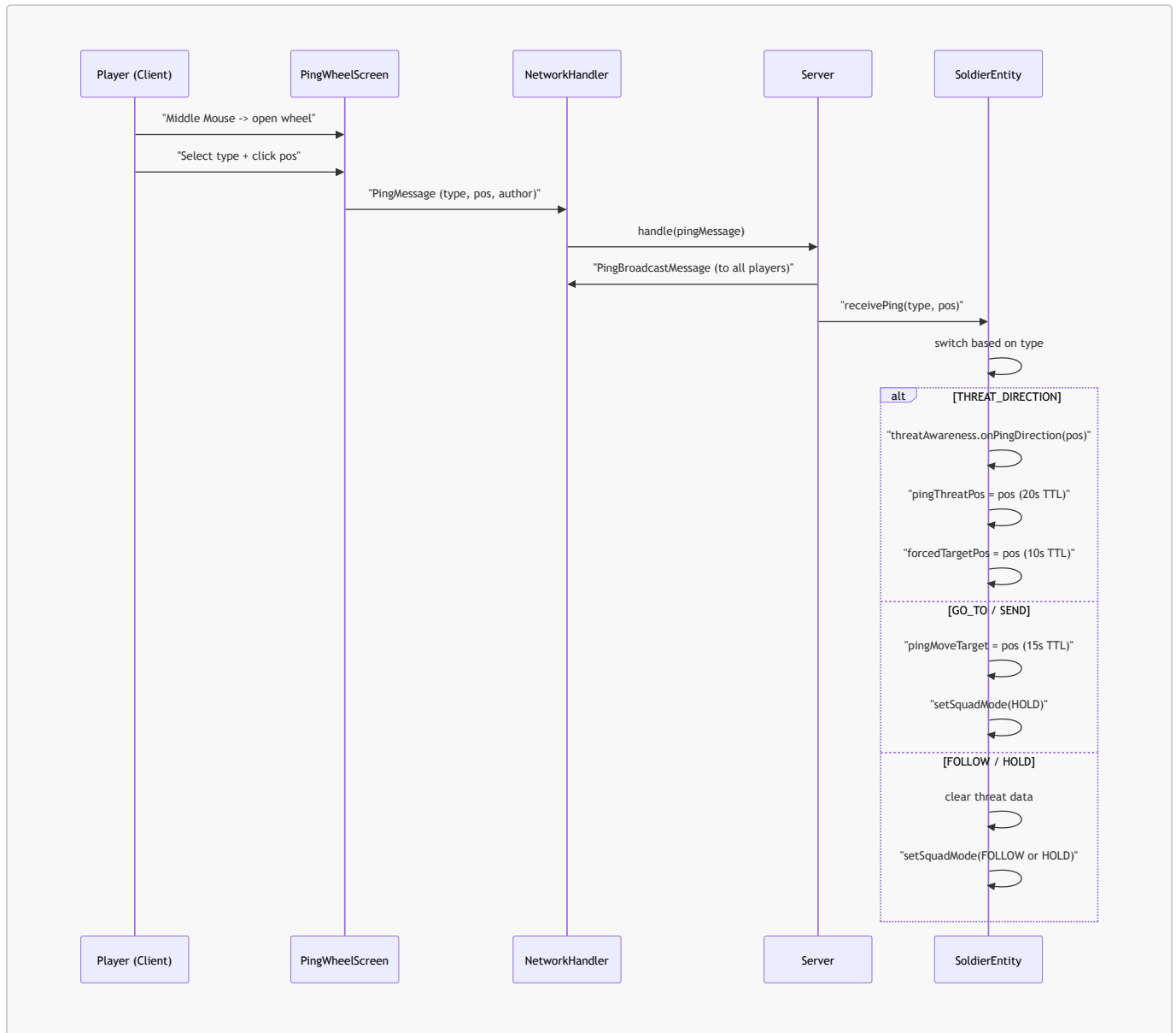
12. Ping System

Package: `com.stevesarmy.ping`

Ping Types

Type	Color	Behavior on Soldier
SEND	#55AAAA	Sets hold position, clears cover, dispatches soldier
GO_TO	#55FF55	Sets ping move target, switch to HOLD
THREAT_DIRECTION	#FF8800	Sets threat direction + forced target pos
LOCATION	#5555FF	No-op (visual only)
FOLLOW	#55FFFF	Switch to FOLLOW, clear all threats
HOLD	#FFAA00	Switch to HOLD at current pos, clear threats

Ping Flow



PingManager (Client-side)

- Stores active pings with 7-second lifetime
- Updates screen positions each frame via `MathUtils.worldToScreen()`
- Scale formula: $2.0 / \text{pow}(\text{distance}, 0.3)$, clamped to [0.5, 2.0]

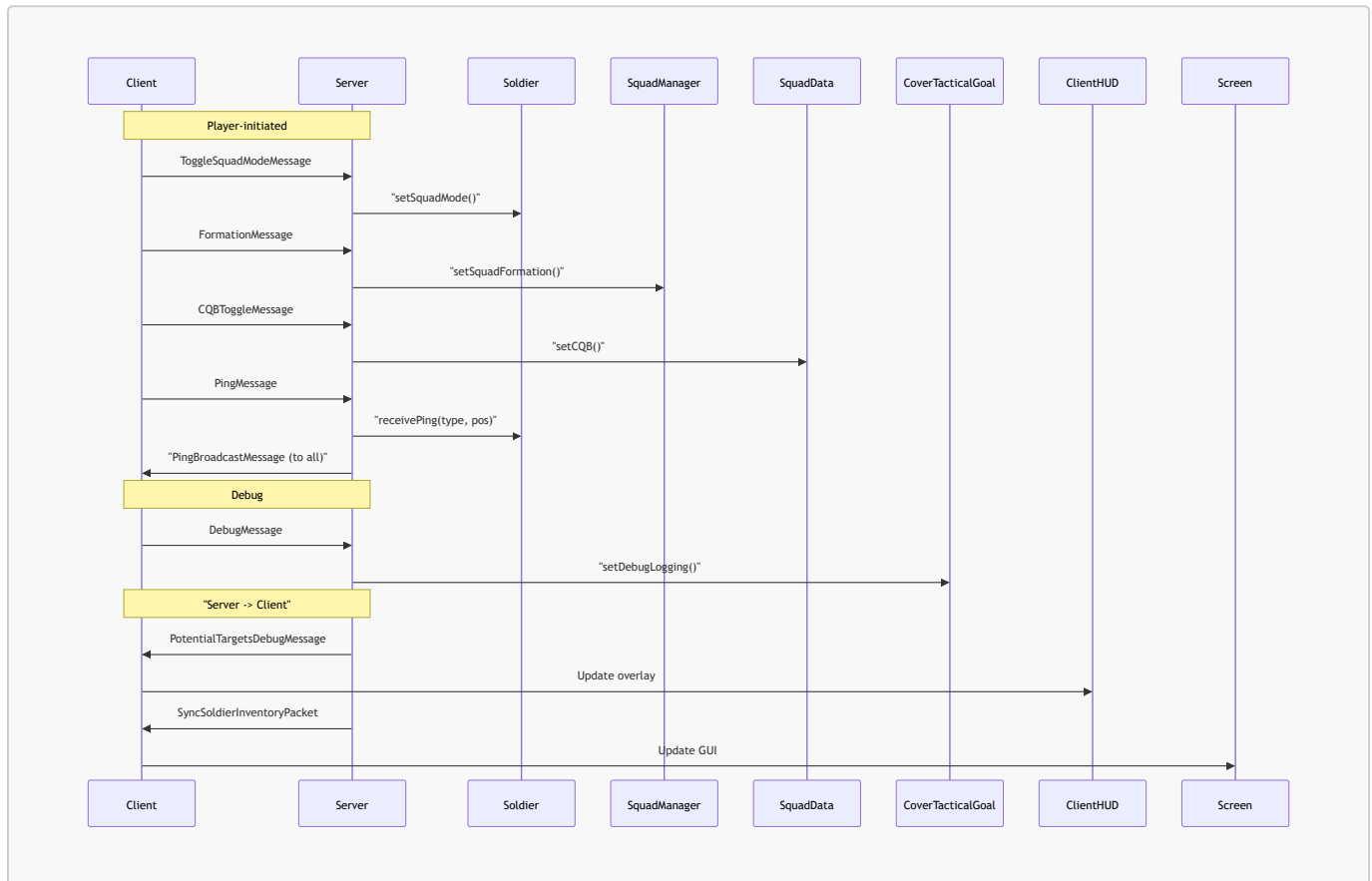
13. Network Layer

Package: `com.stevesarmy.network`

Packet Registry

ID	Packet	Direction	Purpose
0	<code>ToggleSquadModeMessage</code>	C->S	Toggle soldier FOLLOW/HOLD mode
1	<code>DebugMessage</code>	C->S	Toggle debug logging
2	<code>OpenSoldierInventoryMessage</code>	C->S	Request to open soldier inventory
3	<code>PingMessage</code>	C->S	Player sends a ping

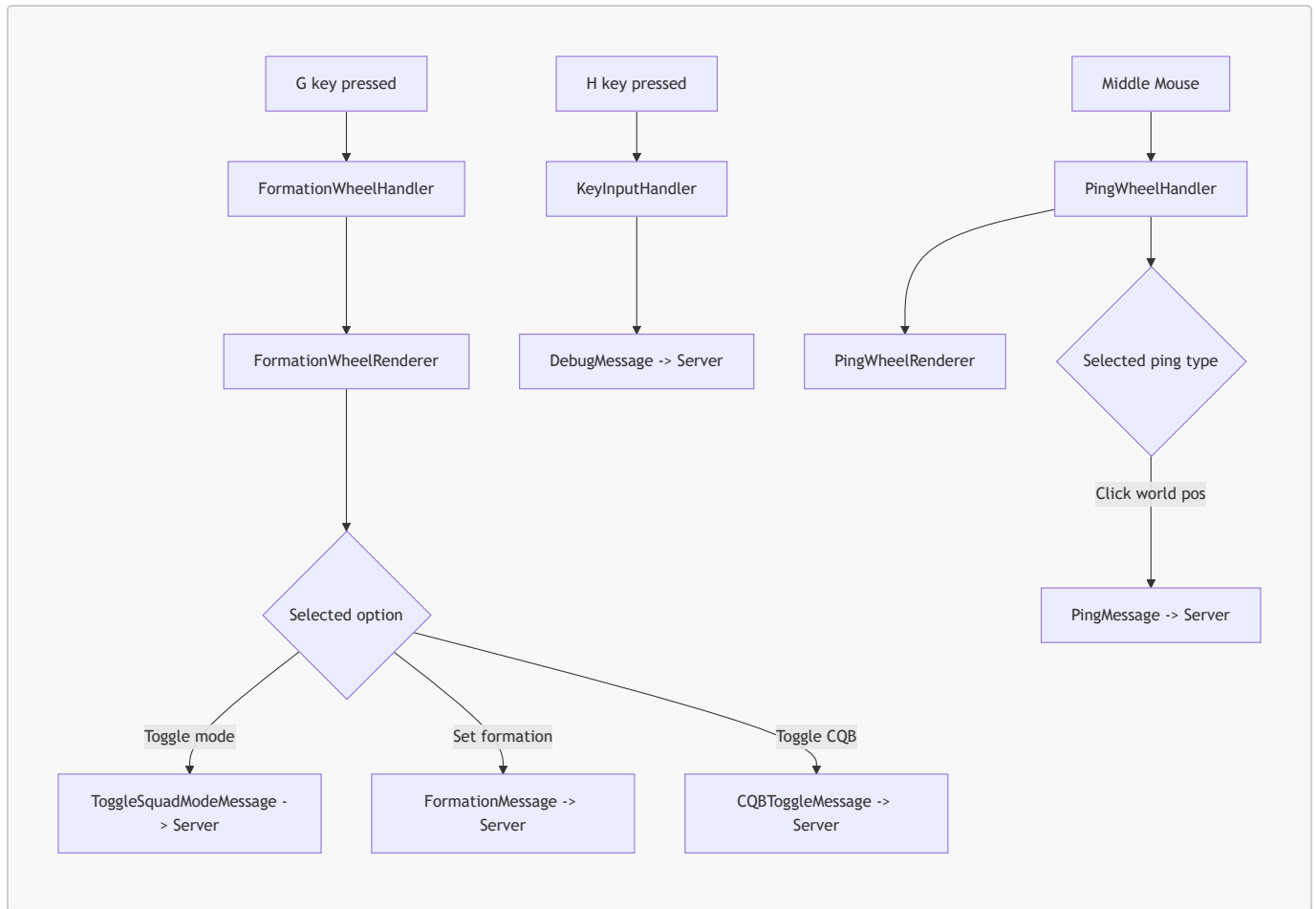
ID	Packet	Direction	Purpose
4	PingBroadcastMessage	S->C	Broadcast ping to all players
5	PotentialTargetsDebugMessage	S->C	Debug data for HUD
6	FormationMessage	C->S	Set squad formation
7	SyncSoldierInventoryPacket	S->C	Sync inventory to client for screen
8	CQBToggleMessage	C->S	Toggle CQB mode for squad



14. Key Bindings & Client

Package: `com.stevesarmy.client`

Key	Mapping	Handler	Action
G	FORMATION_WHEEL	ClientInputHandler	Opens formation wheel (squad mode selector)
H	DEBUG	ClientInputHandler	Toggles debug HUD overlay
Middle Mouse	PING_WHEEL	PingWheelHandler	Opens ping radial wheel

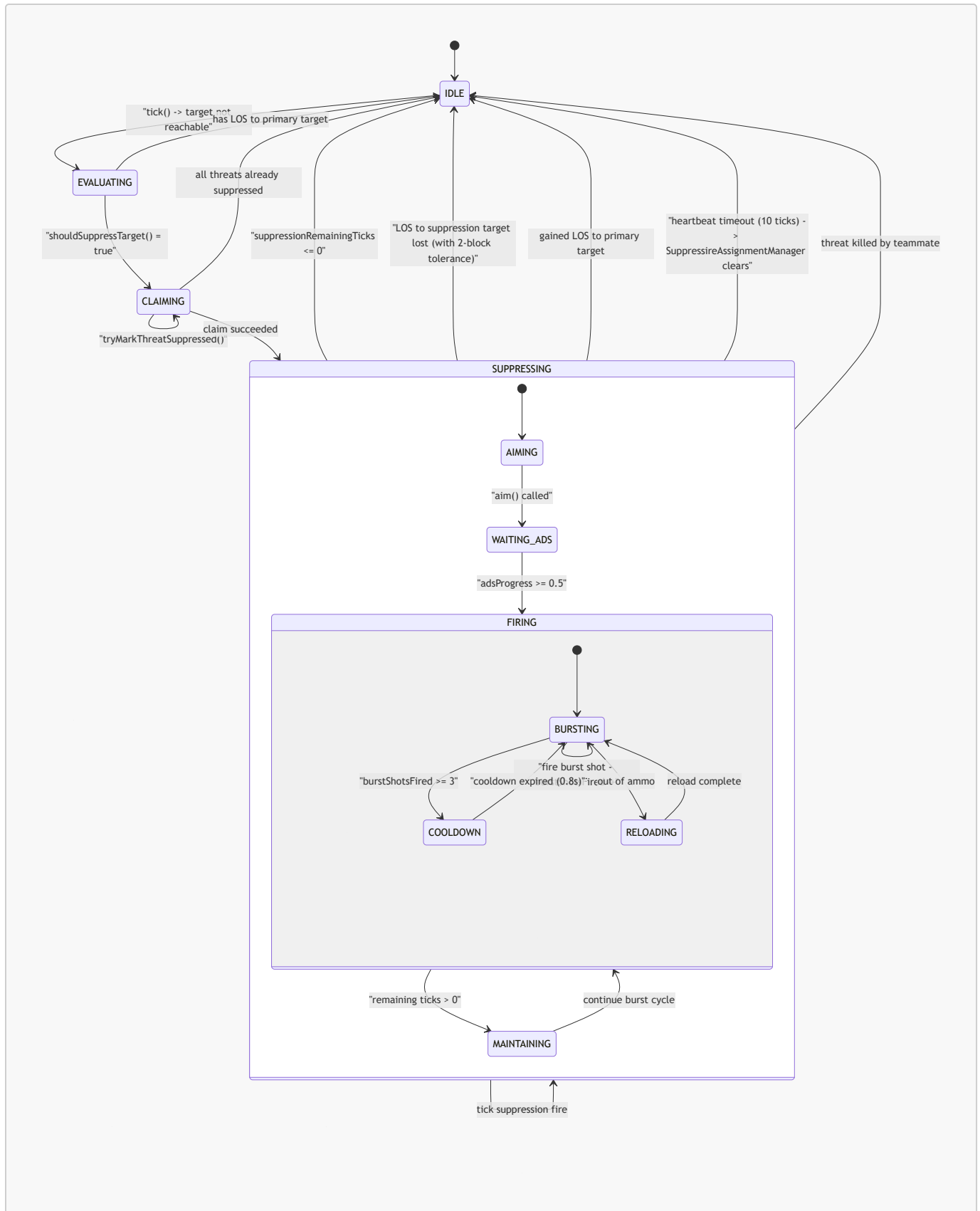


Client Renderers

Renderer	Purpose
CombatDebugRenderer	Target lines, detection arcs, locked target markers
CoverDebugRenderrer	Cover points, scoring, peek candidates
CoverHudRenderer	Soldier cover state overlay
PingOverlayRenderer	World-space ping markers
PingWheelRenderer	Radial wheel UI
FormationWheelRenderer	Squad mode/formation radial UI

15. Suppression Fire Lifecycle

This section ties together `SoldierCombatGoal`, `SquadThreatIntel`, and `SuppressireAssignmentManager`.



Suppression Fire Constants

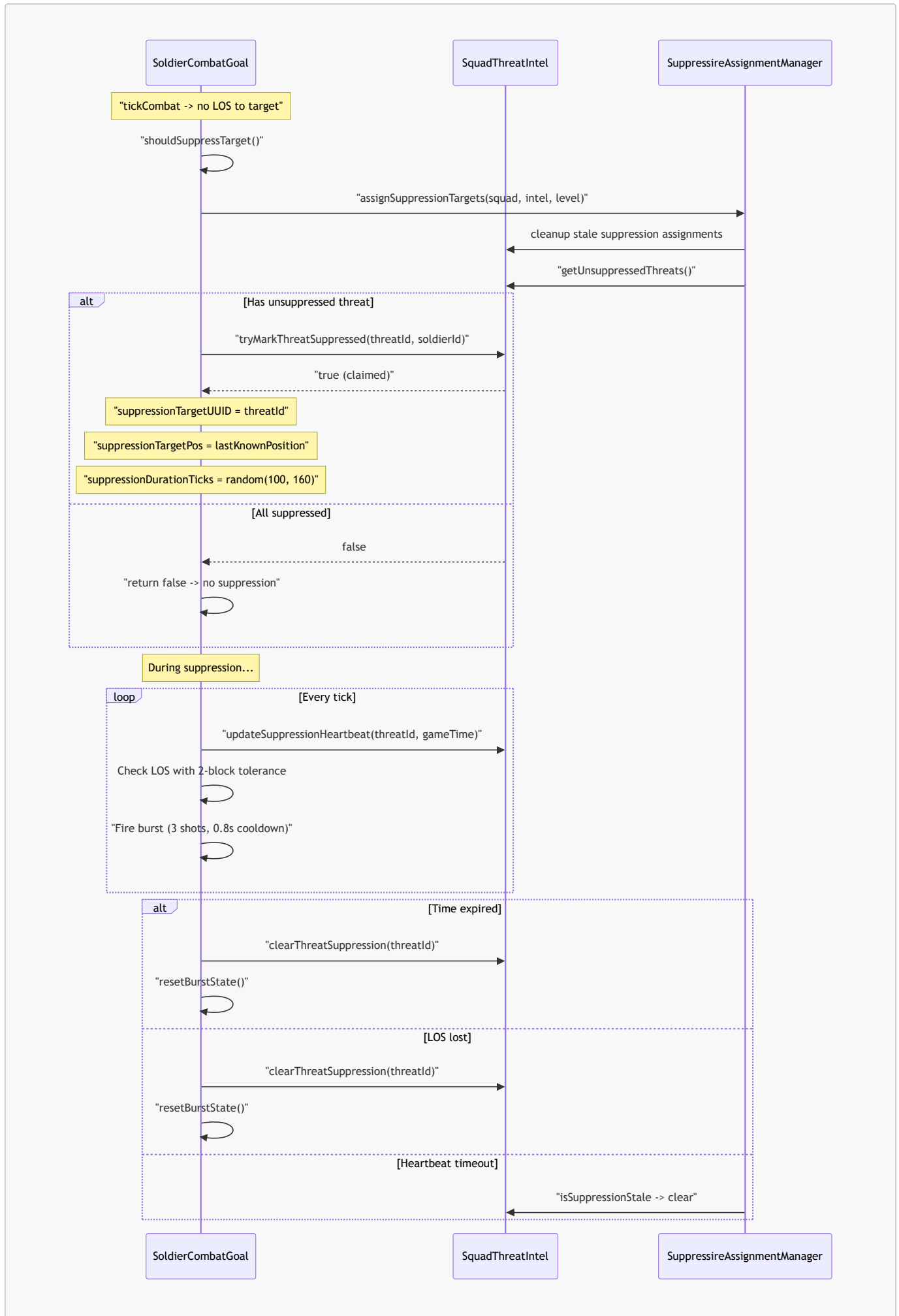
Constant	Value	Purpose
SUPPRESSION_ADS_THRESHOLD	0.5	Minimum ADS progress to fire
SUPPRESSION_MAX_RANGE	30.0	Max distance to suppression target
SUPPRESSION_LOS_TOLERANCE	2.0	Blocks of tolerance for LOS check

Constant	Value	Purpose
SUPPRESSION_MIN_DURATION_TICKS	100	5 seconds min suppression
SUPPRESSION_MAX_DURATION_TICKS	160	8 seconds max suppression
BURST_SHOTS_TARGET	3	Shots per burst
BURST_INTERVAL_SECONDS	0.8	Cooldown between bursts
STALE_TIMEOUT_TICKS	60	3 seconds -- threat stale check

Suppression Spread Calculation

```
spreadRadius = distance x tan(aimInaccuracy_rad)
offset = random(-spreadRadius, +spreadRadius) x 2.0
target += Vec3(offsetX, offsetY x 0.5 + 1.0, offsetZ)
```

Claim/Release Flow



Cross-Module Interaction Summary

